

DEVELOPMENT THROUGH DIGITAL MEDIA

School of Journalism and New Media Studies
Indira Gandhi National Open University
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BLOCK 3 & 4

Unit 11	Prof. K.S. Arul Selvan, Director & Professor SOJNMS	Unit 12 & 16	Dr. Geetika Vashishata, Adjunct Faculty, Manipal Institute of Communication, MAHE, Manipal
Unit 13	Dr. Chhavi Garg, Assistant Professor, Dept. of Journalism and Mass Communication, Central University of Punjab	Unit 14	Adopted from MR-011, IGNOU
Unit 15	Dr. Sujeet Kumar, Assistant Professor, Dept. of Journalism and Mass Communication, Central University of South Bihar, Gaya	Unit 17	Dr. Anusuah Rajendran, Assistant Professor, Dept. of Visual Communication, Mother Teresa Women's University Kodaikanal, Tamil Nadu

PROGRAMME COORDINATORS: Dr. Ramesh Yadav and Prof. K.S. Arul Selvan

COURSE COORDINATOR: Prof. K.S. Arul Selvan

Secretarial Assistance: Ms. Sunita Rawat

Print Production: Mr. Sanjay Aggarwal, Assistant Registrar, MPDD, IGNOU, New Delhi

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Block-1

Internet of Social Scenario

THE PEOPLE'S
UNIVERSITY

BLOCK 1: INTERNET OF SOCIAL SCENARIO

Introduction

Unit 1: Information Society deals with the emerging knowledge society's meaning, features, and basis. One of the basic features of the current pace of globalisation is its primary importance of ICTs, and the emergence of ICT led knowledge society. Here, knowledge becomes a commodity, and it becomes essential to possess the knowledge to achieve success in life and accumulate wealth.

Unit 2: Emerging Trends - Media, Internet, Globalisation explores the role of Information and Communication Technology-based media systems in the post-Cold-era period, the requirements of digital literacy for the ordinary people to be part of the technology-enabled platform and how international human rights movements utilises the ICT platforms for their mobilisation efforts.

Unit 3: ICTs and Women (Issues of Access and Equity) discuss the implications of gender disparity in access to ICT resources and the digital inequality that exists between various gender, particular references to women. Equally this Unit, explores the potential benefits of ICT for women, and incidentally, Sustainable Development Goal 5 focuses on women empowerment through ICT.

Unit 4: Open Source Movement deals with the need and necessity for the proliferation of alternative and open source software initiatives across the computing platforms and explores the characteristics and features of this alternative software and its application in the contemporary scenario.

Unit 5: New Media and Ethical Issues deals with our personal and public conduct in the online space. It explains how new media needs to be used with care without breaching any law or violating the online rights of other users.

UNIT 1 INFORMATION SOCIETY

Structure

- 1.0 Introduction
- 1.1 Learning Outcomes
- 1.2 Technological Transformation and Human Progress
- 1.3 The Emergence of Information Society
- 1.4 What is a Knowledge/Information Society?
- 1.5 Knowledge Economy and Knowledge Workers in an Information Society
- 1.6 Skill Acquisition and Training for Work in Knowledge Society
- 1.7 ICT Infrastructure and Knowledge Dissemination
- 1.8 Let Us Sum Up
- 1.9 Further Readings
- 1.10 Check Your Progress: Possible Answer

1.0 INTRODUCTION

The forces of globalisation redefined contemporary human life's economic, social, political, cultural, etc. They were revolutionising developments in Information Technologies (ITs) which occurred during post World War II or, more specifically, in the 1970s and afterwards, was an integral part of the globalisation process that picked up increased momentum during this period. The rapid interaction and interconnectedness between and among societies created by the current phase of globalisation left Information and Communication Technologies (ICTs) dominant in every aspect of the social system, where information processing and communication technologies became the core of productivity. Initiation in information handling, transmission, storage and retrieval becomes the key to human programmes and development and qualitatively different ways of life. In the emerging society—the information/ knowledge society— knowledge and information, the application of knowledge and information to knowledge generation and information processing/ communication became the fundamental constituents of human progress. It paved the way for the emergence of a global knowledge economy – a networked society with varied economic and educational requirements and principles of organising society, its moral values and identity.

It is restructuring the global social and economic equation—shifting from the income divide to the knowledge divide. In developed countries, ICTs have been the drivers of the knowledge society. As evident from the above discussion, the information age knowledge becomes the primary form of

capital, and the accumulation of knowledge drives economic growth. Here the product with a high knowledge component generates higher returns and higher growth potential. In the knowledge economy, as distinct from peasant and industrial economies where economic wealth was produced by using human manual labour and machines respectively, the generation, dissemination and exploitation of knowledge produce economic wealth predominantly. Thus, productivity is based on the acquisition or generation, dissemination, and application of knowledge or information in the emerging knowledge society.

The main objective of this Unit is to try and know more about information society. We will also try to trace its emergence and list its characteristics here. How and why the generation, dissemination and application of information/knowledge become an integral part of information society and dimensions of work participation in information society also will be analysed in this Unit.

1.1 LEARNING OUTCOMES

After reading this Unit, you will be able to:

- identify the factor that fueled the emergence of the information society in the age of ICT;
- discuss the distinctive features of the information society, the information economy and information/knowledge workers;
- explain the process of skill acquisition and knowledge dissemination in the information society; and
- discuss the role of knowledge and ICTs in empowering communities.

1.2 TECHNOLOGICAL TRANSFORMATION AND HUMAN PROGRESS

Technological transformation has always played a crucial role in the progression of human societies from one stage to another. This transformation has widely influenced society's economic, social, cultural and political-institutional arrangements by introducing changes like work participation in the organisation of production. The transformation of human societies from pre-industrial/agricultural to industrial and then again to post-industrial has widely been shaped by the innovation of new technologies. At the beginning of the 19th century, far-reaching changes in the social and economic lives of humanity were ushered by science, engineering and technology. The changes of that era were marked by the concerted efforts to abolish slavery and large-scale expansion of centralised factory production and the creation of industrial classes — workers and capitalists. This was characterised by the production of manufactured goods and acquiring new skills required for industrial manufacturing. The latter half of the 20th century witnessed the advent of Information and Communication Technologies (ICTs), which heralds a new phase in history and the changes brought about by it in the social and

economic fabric are effectively unique. There was a phenomenal expansion of computer communication, electronic technology, and the service economy during this period. Over time, change has also been marked in the pattern of work participation. In the wake of industrialisation and rapid urbanisation, there has been a shift in work participation from agricultural to nonagricultural economies in the developed and developing countries of the world. However, this shift has taken a new turn in the wake of the emergence of the post-industrial society whereby work participation increased in the service economy, including those in telecommunications, transport and marketing. It is significant that till the early decades of the last century, a large segment of the workers of the industrialised nations like those of France, the United Kingdom, America, Belgium, Japan etc., were in agriculture.

Although agriculture accommodates a substantial proportion of the workforce, there has been an increasing contribution of the service sector to the GDP in developed and developing countries. The blue-collar worker emerged very fast from the last quarter of the 19th century, and then growth became very fast till the second half of the 20th century. Indeed the industrial workers grew phenomenally in the first half of the last century in factories, mines, and transportation, and by the 1950s, they emerged to be the substantial majority of the working population in the industrialised countries. However, in the latter part of the 20th century, they have declined equally rapidly, first as a proportion of the total and since the early 1980s, even in absolute numbers. The emergence of the service sector (we will learn more about this in the later part of this Unit) as a potential avenue for employment and earning has paved the way for the emergence of the knowledge economy both in developing and developed countries. According to Alvin Toffler, agricultural waves, industrial waves and the information age are the three stages of the economic evolution of humanity. Presently, human society is

Box 1.1: Toffler's First, Second and Third Wave

In his book *The Third Wave*, Toffler describes three types of societies, based on the concept of 'waves' - each Wave pushes the older societies and cultures aside.

- After the agrarian revolution, First Wave was the society and replaced the first hunter-gatherer cultures.
- The main components of the Second Wave society are the nuclear family, factory-type education system and the corporation. Toffler writes: "The Second Wave Society is industrial and based on mass production, mass distribution, mass consumption, mass education, mass media, mass recreation, mass entertainment, and weapons of mass destruction. You combine those things with standardisation, centralisation, concentration, and synchronisation, and you wind up with a style of organisation we call bureaucracy."
- The Third Wave is the post-industrial society. Toffler would also add that since the late 1950s, most countries are moving away from a Second Wave Society into what he would call a Third Wave Society. He coined many words to describe it and mentioned names invented by other people, like the Information Age.

Source: The Third Wave 1980

undergoing the third Wave, i.e. the information wave, which is marketed by exposure to developments in information technologies and the predominance of service employment (Toffler 1980).

Check Your Progress: 1

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Explain Alvin Toffler's three types of societies?

.....
.....
.....

2. What technological innovations fueled the speed of the information revolution?

.....
.....
.....

3. List any five features of the information society?

.....
.....
.....

1.3 THE EMERGENCE OF INFORMATION SOCIETY

The roots of information society ideas are closely associated with post-industrialism. Although the scientific and industrial predecessors of electronics-based information technologies can be found in the late 19th and early 20th centuries, it was during the Second World War and its aftermath that the significant technological breakthrough in electronics took place: the first programmable computer, the transistor, source of microelectronics — the true core of Information Technology Revolution (see Box 1.2) (Castells 1996). Manuel Castells contends that the new information technologies, including microelectronics, computers, and telecommunications, diffused widely in the 1970s, accelerating their synergistic development and converging into a new paradigm.

Box 1.2: Tracing Information Technology Revolution

Although the technological inventions such as the telephone by Bell in 1876, the radio by Marconi in 1898, vacuum tube by De Forest in 1906 were landmark inventions in technological development, major

technological breakthroughs leading to a technological revolution in human history based on electronics-based technologies can be said to happen during and after the Second World War. The invention of the transistor in 1947 made possible the processing of electronic impulses at a fast pace in binary mode of interruption and amplification, thus enabling the coding of the logic and communication with and between machines. These processing devices are semiconductors, which are popularly called chips. A decisive step in microelectronics had taken place with the invention of integrated circuits in 1957. It triggered a technological explosion. The giant leap forward in the diffusion of microelectronics in machines came in 1971 with the invention by an Intel engineer of Silicon valley, Ted Hoff, of the microprocessor, the computer on a chip. Thus, information processing power could be installed everywhere.

A combination of three characteristics evaluates the power of chips: their integration capacity, indicated by the smallest line width in the chip measured in microns (1micron is equal to 1 millionth of an inch); their memory capacity, measured in bits: thousands (k) and millions (megabits); and the speed of the microprocessor measured in megahertz. The technological advancements of the microprocessors were so fast that while the first microprocessor of 1971 laid in lines of about 6.5 microns, and the microprocessor of 1999 measured 0.25 microns [*The Intel Core i9 processor of 2018 onwards is about 14 Nanometer - equal to 0.014 microns*]. Greater miniaturisation, further specialisation and the decreasing price of increasingly powerful chips made it possible to place them in every machine in our everyday life. The advent of the microprocessor in 1971, with the capacity to put a computer on a chip, turned the electronics world and indeed the world itself upside down. The microcomputer or personal computer software also emerged in the mid-1980s out of the enthusiasm generated by two Harvard drop-outs, Bill Gates and Paul Allen. Having realised its potential, they went on to found Microsoft, today's software giant.

Indeed, to advance in microelectronics and software, significant leaps forward in networking capabilities have to be added, which was made possible by substantial developments in telecommunications and computer networking technologies during the 1970s. During this period, telecommunications were also revolutionised by a combination of 'node' technologies (electronic switches and routers) and new linkages (transmission technologies). Major advances in optoelectronics (fibre optics and laser transmission) and digital packet transmission technology dramatically broadened the capacity of transmission lines. Each leap and bound in a specific technological field amplifies the effects of related information technologies.

Source: Castells 1996

Manuel Castells (1996) argues that in the new economy emerging worldwide due to the current phase of the globalisation process, productivity and

competitiveness are largely a function of knowledge generation and information processing or informatisation. In the new information age, knowledge became the power and the tool for capital accumulation. According to Yoneji Masuda (1981), in the post-industrial, information-based society, knowledge, or the production of information values, will be the driving force of society, rather than industrial technologies. Thus in the evolving information age, the generation, dissemination and application of knowledge become the basis of all aspects of information and knowledge. Hence, it is also called information society or knowledge society.

Activity - 1

Do you think the contemporary period of technological transformation plays a significant role in social transformation? Why?

1.4 WHAT IS A KNOWLEDGE / INFORMATION SOCIETY?

According to Daniel Bell, in information (and knowledge) society, science plays an increased role in the productive forces; professional, scientific and technical groups will rise into prominence in addition to the vast expansion of information technology, which includes a converging set of technologies in microelectronics, computing (machines and software), telecommunications/broadcasting, and optoelectronics etc. This will be the axial principle of the economy and society. He forecasts the growth of a new social framework based on telecommunications which may be decisive for how knowledge is created and retrieved and the character of work and occupations people are engaged in. The computer will play a pivotal role. In the information society, knowledge and information will supplant labour and capital (as in the Marxian view) as the central variables of the economy. Here, the information will be treated as a commodity, and information possession will give its owner more power. There will be more and more penetration of information into more traditional areas of agriculture, manufacturing and services. There will be major social changes resulting from the establishment of new telecommunications infrastructure (Bell 1976). New forms of social interaction based on electronic communications devices are replacing older types of social relations. There is more application of IT to overcome the ecological and environmental problems associated with industrialism.

Manual Castells (1996) prefers to call the emerging society an “informational” society where the generation and transformation of information generation have become the fundamental sources of productivity and power. To Scott Lash (1999), in the information society, the source of power is information. Power in the manufacturing age was attached to the property as the mechanical means of production. The information age is attached to intellectual property in the form of patent, copyright, and trademark to be valued for creating profit. There is thus commodification of information and no time for reflection. In this society, however, it is not the

commodification driving the informationalisation but the informationisation that is driving commodification. In this age, inequality is less defined in terms of the relations of production but more by exclusion.

The society that is emerging is a “knowledge society”, one which is characterised by “new structures” of knowledge, methods of dissemination and technology that permits and sustains “unrestricted” access to knowledge and control over it. Thus in the contemporary phase of human society, the proliferation of information technology has led to the emergence of a mass society that produces knowledge and information on a mass scale as the driving force of the economy (Naisbitt 1986). Consequently, there has been the rise of the category of knowledge workers, who are fast replacing both histories’ traditional groups and industrial society groups, which is quickly becoming the centre of gravity of the working population. This group is also becoming the single largest workforce group in the post-industrial society (Drucker 1994).

The social and economic dynamics of the knowledge society are widely shaped by the new forces of production, the influence of the global market and the state. To Antony Giddens (2000), globalisation and knowledge economy are the co- constituents of the global information order. This economy is populated by an active and reflexive citizenry of wired workers, whose knowledge is the principal source of production; they are non-hierarchical in their work environment.

According to Bob Jessop (2003), knowledge can acquire commodity value after entering the labour market, and once it is made artificially scarce, its access depends on the payment of rent. Knowledge can be transformed into a fictitious commodity by transforming it from collective resources (intellectual commons) into intellectual property (e.g. patent, copyright etc.) for revenue generation; subsuming of knowledge production under exploitative class relations and by transforming intellectual labour into wage labour for producing knowledge for the market; and bringing intellectual labour under capitalism control through commoditisation and integration into a networked digitised production and a consumption process controlled by the capital. He foresees the possibilities of monopolies in knowledge and information by embedding them in technology, standards, or legally entrenching intellectual property rights.

It is now recognised that in the wake of the present phase of globalisation, ICTs have paved the way for the emergence of a global knowledge society and economy; a networked society with a varied kind of economic and educational requirements and principles of organising the society, its moral, values and identity. In essence, the ICTs have been juxtaposed to restructuring economic and social institutional arrangements of the knowledge economy of the information age locally and globally. ICTs now offer a challenge to the conventional ways of getting information, knowing and disseminating. Thus this cutting edge technology has been linked to the new discourse of development. In this information age, knowledge is the basic form of capital

and that the accumulation of knowledge drives economic growth. This has emerged as a symbiotic relationship between knowledge economy and ICTs for releasing the creative potential and knowledge embodied in people and harnessing local-global connectivity for the generation of wealth and to widen the market of this economy.

Following are some of the distinctive features of the knowledge society:

- The basis of knowledge-based development in knowledge societies is the generation, dissemination and deployment of knowledge.
- Scientific knowledge is considered as a knowledge society skillset, and the scientific and technical group will rise into prominence.
- The social network in a knowledge society is based on telecommunication technologies.
- The creation and retrieval of knowledge play a decisive role in the organisation of work and occupation. The occupations, which make more and more innovative knowledge, will become predominant in this economy.
- Knowledge/information is treated as a commodity, and knowledge possession gives the owner more power.
- In a knowledge society, inequality is defined in terms of exclusion from knowledge.
- Knowledge is transformed from collective resources (intellectual commons) into the intellectual property for revenue generation in the knowledge society.
- In a knowledge society, the conflict is between minority knowledge workers and the majority of traditional workers.
- The knowledge society will be far more competitive than the earlier societies, as knowledge will be a key competitive factor for career and earning opportunities.
- Knowledge in the knowledge society exists in the specialised application by specialised experts. The central workforce will be the highly specialised people and not the generalists. Here the people who acquire the specialised knowledge will have more scope of mobility than ever. “It demands for the first time in history that people with knowledge take responsibility for making themselves understood by the people who do not have the same knowledge base. It requires that people learn to assimilate into their own work specialised knowledge from other areas and disciplines.”

Activity – 2

Some of the features of information/knowledge society are given in this text. Can you point out some more features?

1.5 KNOWLEDGE ECONOMY AND KNOWLEDGE WORKERS IN A INFORMATION SOCIETY

In the knowledge economy, economic wealth is predominantly produced by using knowledge. Indeed it is an emerging society whose economic base is widely shaped by the generation, dissemination and exploitation of knowledge. The neoclassical economists have emphasised labour and capital as key factors of development. To Paul Romer (1990), knowledge is the third factor of production and long-term growth; it is the basic form of capital and that economic growth is driven by its accumulation. Here we may sum up the following features of the knowledge economy:

- Knowledge is a public good in the knowledge economy, as this becomes an object of extensive use.
- As the knowledge economy depends on the generation of knowledge for its prosperity, knowledge gained by experience is as important as formal education and training.
- A knowledge economy becomes a learning economy to utilise its full capacity and take its optimum advantage. Learning means using new technologies to access global knowledge and communicating with others about innovation. In the learning economy, individuals, firms and countries will create wealth in proportion to their capacity to learn and share innovation (Foray and Lundvall 1996; Lundvall and Johnson 1994). Formal education, too, needs to become less about passing on information and focus more on leading people how to learn. Learning thus becomes a lifelong process in the knowledge economy.
- According to OECD, ICTs are the facilitators of knowledge creation. In the knowledge economy, ICTs release people's creative potential and knowledge. Wealth generation is becoming more closely tied to the capacity to add value using ICT products and services.

Individuals have been put in the centre of the knowledge and skills-based society in the information age. Individuals want to master their own lives and expect to contribute to the economy and society more than ever before. The development of individuals as active citizens of society is increasingly given a central place in statements of learning, education and training objectives.

The individual is becoming the architect and builder responsible for developing his/her skills, supported by public and enterprise investment in lifelong learning. ICTs empower the individual from a passive teacher-oriented approach to gaining knowledge; there is a shift towards learning for

life and work centred around the individual. The need to learn how to access, analyse and exploit information and transform it into new knowledge is increasing, and in particular, Internet-based technologies offer great opportunities. The empowered individuals or the knowledge workers take charge of all spheres of society.

Knowledge workers of the information society are distinctively different from those of the agrarian and industrial society workers. They are defined as “symbolic analysts” who manipulate symbols rather than machines. They include architects and bank workers, fashion designers and pharmaceutical researchers, teachers and policy analysts. They are associated primarily in the service sector such as telecommunications, transport and financial services. Knowledge workers systematically accumulate knowledge, share it and deploy it purposefully. Continuously improving the stock of knowledge will be critical for their success. In the information society, the knowledge workers are valued very highly. For e.g. in many American manufacturing companies, intangible assets are now worth more than tangible assets. These intangible or intellectual assets are based primarily on the skills and capabilities of their so-called knowledge workers.

The distinctive features of the knowledge workers are:

- The knowledge workers are the leading class of the information society and necessarily the ruling class. They differ fundamentally from the other groups in history who occupied the leading dominant position in the values, expectations and social position.
- They get access to work and social positions in information society through formal education and training.
- Quantity and quality of knowledge work will differ substantially based on the amount and kind of formal knowledge and training required for a particular job.
- As formal education occupies the centre stage of the information society, formal schooling emerges to be the key institution. Here, the components of knowledge (knowledge mix), quality of learning, and teaching become a central concern of the knowledge society and central political issues. “In fact it may not be fanciful to anticipate that the acquisition and distribution of formal knowledge will come to occupy the place in the politics of the knowledge society which acquisition and distribution of property and income have occupied in the age of capitalism”.
- Significantly, not necessarily the conventional system of schooling, but the systematic continuing education offered in the place of employment would get importance. Here, an educated person will have learned how to learn and continue to learn throughout her/ his lifetime, especially in and out of formal education. Thus the acquisition of knowledge is not age-specific but lifelong.
- The knowledge workers are to learn different kinds of terms for different purposes – their performance capacities, strengths, limitations and

tradeoffs between various kinds of terms. They are also to learn how to switch from one kind of team to another and integrate themselves into a team.

- Organisations, in general, provide the platform for the knowledge workers to convert their specialised knowledge into performance. In the organisation, the knowledge workers are at times the employee and at times the bosses.
- The knowledge workers also own the tools of production. Unlike the capitalist society, true investment in the information society is the knowledge of the knowledge workers; without knowledge, the whole production process is unproductive. The knowledge investment determines whether the employee is productive or not, rather than the tools, machines, and capital the organisation furnishes.

Activity – 3

What do knowledge workers do? People working in tourism and sector are knowledge workers. Do you agree with this statement? Why?

The Three Levels of Knowledge-Based Development

As we have seen in the previous discussions in the information age, knowledge has a broader meaning. In the past, clever and creative people also used knowledge to design innovative products and services. But in the information age, instead of knowledge being vested in one or two creative people, it will be embedded in systems and databases and made available to all. To achieve maximum effectiveness, knowledge must be systematically accumulated, shared and purposefully deployed.

That means a knowledge-based society is centred on the three processes of knowledge accumulation, knowledge dissemination of the accumulated knowledge and application of that knowledge for the productivity of the society. An analysis of this process of knowledge accumulation, dissemination and deployment in terms of skills, infrastructure and experience concerning knowledge production will enable us to assess the dimensions of knowledge society and economy. It is required to take stock of the literacy and higher education levels to examine the skills for knowledge accumulation. The size and growth of the telecommunication network will echo the infrastructure required for knowledge dissemination, and economic structure will reflect the level of application of the knowledge in a knowledge-based society.

1.6 SKILLS ACQUISITION AND TRAINING FOR WORK IN KNOWLEDGE SOCIETY

From the preceding sections of this Unit, we already gathered that the key characteristic of the knowledge economy lies in the belief that wealth (or productivity) is increasingly dependent on the development and application of new knowledge by specialist knowledge workers. It has been increasingly

recognised that in the knowledge society, people's endowment of skills and capabilities and investment in education and training constitute the key to economic and social development. It is not so much physical capital or human skills (human capital) that determines economic growth. The nation's capability to apply knowledge to knowledge itself is essential to economic development. Economies are increasingly being built on a foundation of information, learning and adaptation. Here both the quantity of knowledge increases and the production of knowledge accelerates (Scott 1997).

So an important aspect of the emergence of the knowledge society is the readiness to acquire new skills. ICT use represents an augmentation of human skills and capabilities. In examining the skills, it is vital to develop measures that indicate the state of readiness to enlarge the use of information to develop knowledge. A principal indicator of such readiness is literacy level. Literacy is the first indicator of attaining the skills level needed for the productive use of ICT – an imperative of the information age. Here literacy means more than knowing how to read, write or calculate. It involves understanding and using the information required to function effectively in the knowledge-based societies that will dominate the twenty-first century.

Illiteracy is a fundamental barrier to participation in knowledge societies. The vast majority of the illiterate population will be excluded from the emerging knowledge societies. The skill attainment is hierarchical. The hierarchy begins with the attainment of basic literacy. All the work processes in which ICTs can contribute to economic growth require basic literacy.

In knowledge societies, it is recognised by governments and organisations that knowledge contributes to individual well being societal and economic growth. This recognition is translated into action when new models for lifelong learning are encouraged. Enterprises can improve productivity and compete successfully in increasingly integrated world markets by investing in their human resources. For e.g. in Denmark, enterprises that introduced the process and product innovation combined with targeted training were more likely to report output growth. Countries with the highest incomes are also those where workers are mostly educated.

Even though higher education has always been formally designed as a structure for producing and organising advanced knowledge, the emergence of a knowledge economy and the importance of globalisation and ICT place new demands on higher education. Firms that wish to compete in the global economy will have to possess the organisational abilities/ knowledge to maintain or increase their competitive advantage in a turbulent market environment. It implies that there is a need for firms to have and/or train a flexible and versatile workforce. Firms, therefore, will express a continuous demand for courses in which their employees are retrained. In other words, great emphasis has been given to lifelong learning and the realisation of learning society. For the education of students, one of the implications of the knowledge-driven economy is that students will have to be prepared for a labour market in which they could change jobs many times during their

working career. This means that students should acquire appropriate skills for this. This will have to be reflected in the higher education curriculum – in its content, structure, length and mode of delivery. Thus, higher education has become a tradable product in the knowledge society.

The developed countries have higher access to ICTs than the developing countries. The fast proliferation of ICTs in developing countries is widely due to sustained investment in education, research and development activities. Similarly, the developed countries have been consistently spending a higher proportion of their public expenditure on higher education. Advanced countries invest at least 30 times more per student in education and training than in the LDCs. However, the developing countries started spending more on education than was spent previously. It becomes evident that human resources development and training contribute to improved productivity in the economy, reduce skills mismatches in the labour market, and promotes international competitiveness.

Another important consequence of the acceleration of scientific and technological progress is the diminished emphasis on remembering countless facts and basic data and the growing importance of methodological knowledge and analytical skills — the skills needed to think and analyse information autonomously. Today, in several scientific disciplines, elements of factual knowledge taught in the first year of study may become obsolete before graduation. The learning process needs to be increasingly based on finding and accessing knowledge and applying it in problem-solving. Learning to learn, transforming information into new knowledge, and translating new knowledge into applications become more important than memorising specific information. In this new paradigm, priority is given to analytical skills: the ability to seek and find information, crystallise issues, formulate testable hypotheses, marshal and evaluate evidence, and solve problems. The new competencies that employers value in the knowledge economy include oral and written communications, teamwork, peer teaching, creativity, envisioning skills, resourcefulness, and the ability to adjust to change.

Lifelong Learning: The second dimension of change in education and training needs is the short “shelf life” of knowledge, skills, and occupations and, as a consequence, the growing importance of continuing education and regular updating of individual capacities and qualifications (Wagner 1999). In OECD countries, a lifelong-education model progressively replaces the traditional approach of studying for a discrete and finite period to acquire a first degree after secondary school or complete graduate education before moving on to professional life. Graduates will be increasingly expected to return periodically to tertiary education institutions to acquire, learn to use, and relearn the knowledge and skills needed throughout their professional lives. This phenomenon goes beyond the narrow notion of a “second chance” for out-of-school young adults who did not have the opportunity to complete much formal study. It has more to do with the updating and upgrading of learning required to refresh and enhance individual qualifications and keep pace with innovations in products and services. The concept of “lifelong

learning for all” adopted in 1996 by the OECD ministers of education stems from a new vision of education and training policies supporting knowledge-based development. Lifelong learning requirements may lead to a progressive blurring between initial and continuing studies.

Activity – 4

What is the significance of lifelong learning in knowledge societies?

1.7 ICT INFRASTRUCTURE AND KNOWLEDGE DISSEMINATION

In knowledge societies, knowledge creation is important; its dissemination and knowledge sharing with the world around it are equally important. In the information age, ICTs are the main medium for knowledge dissemination. In this information age, the info-technological revolution is restructuring the global social, economic equations — shifting from the income divide to the knowledge divide. Amidst the contemporary information revolution, which ushers with it a world of greater interconnectivity, accelerating the flow of data and shrinking time and national boundaries. Accessibility of the World Wide Web (WWW) is turning the world into a global village. With the decreased cost of processing and dissemination of information and increased convergence of information, computer and telecommunication technologies became the base of knowledge societies.

Knowledge sharing is the interactive process of making the right information available to people at the right time understandable to enable them to act judiciously- enriching the knowledge base in the entire mechanism. Knowledge sharing can occur at all levels— between countries, within a country, between communities and among individuals. It can happen from local to global, from poor to rich and vice versa. Knowledge dissemination and sharing became indispensable in today’s life for good governance, participation of people in their development etc. Unrestricted and continuous sharing of global and local knowledge between policymakers, public and private sectors and civil society heralds the way forward to an empowered knowledge society, which can efficiently manage the development change process. It ensures the inclusion of poor and marginalised communities in the change process.

Since the Second World War, rapid technological advances have occurred due to the convergence of telecommunications and computing technology, known as Information and Communication Technologies (ICT). ICT has been the driver of the knowledge society. They provide new and faster ways of delivering and accessing information, innovative ways for real- time communication and new ways to do business and create livelihood opportunities. For ages, knowledge has been passed on from one generation to another through written text, folklore, word of mouth, religions and customs.

The knowledge, however, remained preserved geographically and hierarchically. On the other hand, ICT breaks all the natural, social, cultural and hierarchical barriers to knowledge sharing. It can help the people to leapfrog some of the traditional barriers to development by using knowledge in various ways, such as improving access to information, expanding their market base, enhancing employment opportunities, making government services work better, etc.

In the contemporary global context, the use of ICTs is expanding rapidly. ICTs comprise a diverse set of technological tools and resources to create, disseminate, store and manage data and information. Traditional ICT tools such as television, radio and the telephone have proven their effectiveness in promoting development. The emergence of computers, the Internet and wireless communications technology, and powerful software for processing and integrating text, sound and video into electronic media comprise modern ICT. For the past two decades, the spread of the global electronic network of computers, popularly referred to as the Internet, and wireless telephony has generated an unprecedented global flow of information, products, people, capital and ideas. Internet boundless potential to reach everyone who is connected to target specific information.

The most significant advantage of ICTs is the reach and low cost of technology and data transmission. Technically, every individual can have private or public access to a data terminal, which connects him to each and every individual in the world. Knowledge dissemination and knowledge sharing in knowledge societies depend on ICT infrastructure, mainly telecommunications, computer-mediated communication — the Internet and mass media of communication.

Telecommunication Network

Telecommunication networks are a key facilitator to the knowledge society. Telecommunications systems are one of the most complex systems ever built by humankind. It has penetrated every aspect of human life. In the 19th century, the invention of the telegraph and the telephone forever changed how messages moved worldwide.

Telephony and mobile phones made possible virtually instantaneous two-way communication between any two places on the planet connected by appropriate wiring and switching devices. Most telephony networks were developed as public monopolies, though the US was an exception. Extensive international organisational arrangements were established to ensure interconnectivity through common networks standards. The International Telecommunications Union (ITU) and related Treaty arrangements represent some of the first attempts to develop effective forms of international governance (Wiesman 1998). Since the 1980s, governments have come under increasing pressure to commercialise, privatise and deregulate their telecommunications industries. By the late 1990s, virtually all national telephone networks had partly privatised and opened up to the national and

international competition. This resulted in drastic decreases in international communications services' price, thereby promoting faster and cheaper knowledge dissemination.

Telecommunications is now, but one form of processing information; transmission and linkage technologies are increasingly diversified and integrated into computer-operated networks. Mobile phones show a convergence of different communication technologies. Sending images, text messages and, of course, audio and video are just click-aways for millions of mobile users. Every month, it seems, a new mobile phone comes out that's "smarter" than the last in its ability to gather and transmit a growing amount of data: voice, images, news and more. The technologies of photography, broadcasting, audio system and the Internet all converged into one gadget of mobile phones.

Computer-Mediated Communication Network — the Internet

The Internet network began in the 1960s (see Box 1.3) in the United States and soon became common. The Internet network became the backbone of computer-mediated communication in the 1990s since it gradually links up to most networks. In the mid-1990s, it connected 44000 computer networks and about 3.2 million host computers worldwide with an estimated 25 million users, and it is expanding rapidly (Castells 1998). In the year 2005, the Internet network crossed 6 million computer networks. In 2020, ICT connected 4.5 billion people to the Internet globally. A detailed analysis of internet access is already explained with Unit 3 of the course MNM011, and learners may refer to it.

The rapid evolution of microprocessor technology since its discovery and the swift advances in fibre optic network technologies resulted in the rapid growth of computing power and the communication power of people worldwide. These advances in technology- enabled new types of services to be used in digital format. Technological advances have also slashed the costs of information and communication. Internet telephony offers much cheaper long-distance communication than the traditional telephone. The cost of transmitting digital information anywhere in the world has also fallen dramatically. Until the early 1980s, communication was generally restricted to analogue signalling, which means each telecommunication network was designed to carry different types of information separately. Voice traffic was carried over the telephone system, text used a separate telex network, and high-frequency broadcast networks were dedicated to sending video and audio signals. With digital communication, these separate networks are becoming less differentiated.

Box 1.3: The Beginning of the Internet

The Internet originated in a daring scheme imagined in the 1960s by the technological warriors of the US Defence Department Advanced Research Project Agency (DARPA) to prevent a Soviet takeover or destruction of

American communication in case of nuclear war. To some extent, it was the electronic equivalent of the Maoist tactics of dispersal of guerrilla forces around a vast territory to counter an enemy's might with versatility and knowledge of the terrain. The outcome was a network architecture that, as its inventors wanted, cannot be controlled from any centre and is made up of thousands of autonomous computer networks that have innumerable ways to link up going around electronic barriers. Ultimately ARPANET, the network set up by the US Defence Department, became the foundation of the global, horizontal communication networks.

Source: Castells 1998

The Internet currently carries a combination of pictures, drawings, moving images, sound and text. The technologies of mobile and television, the radio and camera, the fax and word processor, the database and the spreadsheet all are integrated into one system, the Internet, which makes the Internet unique in its capacity to support two-way interactions. Since the early 1990s, the World Wide Web (WWW) has become the mainstream environment for creating and disseminating digital information.

Previously access to the Internet was almost exclusively from personal computers. This has been changing for one decade, with the advent of smartphones. Major online services are available through mobile apps rather than the Internet. This development enabled remote access to the Internet and its related services without a basic ICT infrastructure.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of the unit.

1. Explain the characteristics of information/knowledge society?

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2. Enumerate the profile of knowledge workers?

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3. Discuss the challenges and opportunities of Information Society?

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1.8 LET US SUM UP

During the second half of the 19th century, there began a great revolution in the storage and communication of information. After industrialisation, society moved towards a post-industrial information age where production dissemination and deployment of knowledge became the basis of productivity and social advancement. The evolving information/knowledge societies are marked by rapid advances in science and technology, the convergence of information, computer and communication technologies, the reduced cost of processing and disseminating information, and the increasing connectedness of nations. These revolutionary changes are said to transform societies into smart communities largely through the converging new information, computer and telecommunication technologies (ICTT).

This Unit examined the background of the information revolution and the characteristic features of the emerging society. It also analysed why and how knowledge becomes the basic constituent of this society.

1.9 FURTHER READING

1. Castells, M. (2011). The rise of the network society (Vol. 12). John Wiley & sons.
2. Castells, M., & Cardoso, G. (2005). The network society: From knowledge to policy (pp. 3-21). Washington, DC: Center for Transatlantic Relations, Paul H. Nitze School of Advanced International Studies, Johns Hopkins University. <https://www.dhi.ac.uk/san/waysofbeing/data/communication-zangana-castells-2006.pdf>
3. Lyon, D. (2013). The information society: Issues and illusions. John Wiley & Sons.
4. Van Dijk, J. (2020). The network society. Sage.
5. Webster, F. (2014). Theories of the information society. Routledge.

1.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. In his book The Third Wave, Alvin Toffler describes three types of societies, based on the concept of 'waves' - The first Wave revolves around agriculture-based society; Second Wave characterised industry society, and the Third Wave is the post-industrial society.
2. Advances in microelectronics and software have led to major leaps in networking capabilities, which was made possible by major telecommunication and computer networking technologies developments. Major advances in optoelectronics and digital packet

transmission technology dramatically broadened the capacity of transmission lines.

3. The basis of knowledge-based development in knowledge societies is the generation, dissemination and deployment of knowledge.
 - a. Scientific knowledge is considered an asset in the knowledge society, and the scientific and technical group will rise into prominence.
 - b. In the knowledge society, inequality is defined in terms of exclusion from knowledge.
 - c. The conflict is between minority knowledge workers and the majority of traditional workers.
 - d. Knowledge will be a key competitive factor for career and earning opportunities.

Check Your Progress: 2

1. **Following are some of the distinctive features of the information/knowledge society:** The basis of knowledge-based development in knowledge societies is the generation, dissemination and deployment of knowledge. Scientific knowledge is considered as a knowledge society skillset, and the scientific and technical group will rise into prominence. The social network in a knowledge society is based on telecommunication technologies. The creation and retrieval of knowledge play a decisive role in the organisation of work and occupation. The occupations, which make more and more innovative knowledge, will become predominant in this economy. Knowledge/information is treated as a commodity, and knowledge possession gives the owner more power. In a knowledge society, inequality is defined in terms of exclusion from knowledge.
2. **The distinctive features of the knowledge workers are:** The knowledge workers are the leading class of the information society and necessarily the ruling class. They get access to work and social positions in information society through formal education and training. The knowledge workers are to learn different kinds of terms for different purposes – their performance capacities, strengths, limitations and tradeoffs between various kinds of terms. They are also to learn how to switch from one kind of team to another and integrate themselves into a team. Organisations, in general, provide the platform for the knowledge workers to convert their specialised knowledge into performance. In the organisation, the knowledge workers are at times the employee and at times the bosses. The knowledge workers also own the tools of production.
3. **The opportunities in the information society are:** The most significant advantage of ICTs is the reach and low cost of technology and data

transmission. Technically, every individual can have private or public access to a data terminal, which connects him to each and every individual in the world. Knowledge dissemination and knowledge sharing in knowledge societies depend on ICT infrastructure, mainly telecommunications, computer-mediated communication — the Internet and mass media of communication. The challenges are access to ICTs and information, the ability to utilise the retrieved information in a productive manner and the capability to participate in the wider communication process.



UNIT 2 EMERGING TRENDS - MEDIA, INTERNET, GLOBALISATION

Structure

- 2.0 Introduction
- 2.1 Learning Outcomes
- 2.2 Media
 - 2.2.1 Access to Media Information and Human Rights Protection
 - 2.2.2 The Media as a Tool of Human Rights Abuse
- 2.3 Internet
 - 2.3.1 Internet and Human Rights
- 2.4 Globalisation and Human Rights
 - 2.4.1 Global Social Movements
 - 2.4.2 Globalisation in Media
 - 2.4.3 Globalisation Trade and Investment
- 2.5 Let Us Sum Up
- 2.6 Further Readings
- 2.7 Check Your Progress: Possible Answers

2.0 INTRODUCTION

In the present age of globalisation and the ever-growing role of information technology, both opportunities and challenges have emerged for the promotion and protection of Human Rights. It is well known that the end of the cold war brought major changes in international political and economic relations. In the process, a new world economic order guided by international agencies like the International Monetary Fund, World Bank, World Trade Organisation, etc., has emerged. This has greatly removed trade and tariff barriers, making Transnational companies' entry into many countries quite easy. A key component of the post- Cold War-era since 1991 information flows, rapidity of transmission and transparency characterised, among other things, the contemporary world. Modern communications, especially the Internet and television, have rendered time-space factors obsolete. Over the years, news anchors, reporters, photojournalists, and other media actors in humanitarian crises have become a common occurrence. In this sense, the media has emerged as a powerful instrument in bringing to light violations of human rights. States are under pressure to fulfil international obligations while also losing some of their sovereign powers. Thus there have emerged interesting developments and challenges for the human rights movement worldwide. It is true for India also. In this Unit, we will discuss some of these developments.

2.1 LEARNING OUTCOMES

This Unit deals with the emerging trends in the Human Rights movement in India. After going through this Unit, you will be able to:

- understand the place of media in the post-Cold war era and global public scrutiny;
- explain the nature of the Internet and the challenge of digital literacy for the 21st century; and
- evaluate how the international human rights movement is an integral part of globalisation.

2.2 MEDIA

The traditional wisdom would have us believe that media and democracy reinforce, sustain and nourish each other. The former is considered the Fourth Estate, the other three pillars being the organs of the government, namely, the legislature, executive, and the judiciary. According to Rajni Kothari, modernisation, conceived as adherence to the principles of the rule of law, equity, justice, freedom, equality and other related principles, is facilitated by mass media. The Indian mass media played this stellar role during the freedom struggle. In his pioneering work, *State Against Democracy: In Search of Humane Governance*, Kothari reminds the state of its bounden duty to respect and protect people's fundamental freedoms. The media have a right and a duty, as a check on the state's power to inform, to report truthfully about government misconduct and to expose abuse of public authority. As a derivative of the latter, power tends to corrupt its wielders, and absolute power corrupts absolutely.

2.2.1 Access to Media Information and Human Rights Protection

With a globalised media, states are now confronted with the glare of global public scrutiny into their human rights practices. A state that violates human rights norms, accepted by the global community in various international treaties, may attract the opprobrium of other countries and international organisations of international audiences and its citizens. Besides, global modes of communication can inform local populations about the abuses their governments are veiling from them, thus inducing the formation of social or civil movements against the status quo. The global sharing of information can not only provide victimised groups access to supporters throughout the world, but it can also increase the costs to the state for its repressive domestic policies towards its citizens. As the ability to access information increases, the dispersal of knowledge becomes virtually impossible to stop. This circulation of information impels the estate to act transparently. Underlying those described above, the Vienna World Conference on Human Rights (1993) stressed the importance of the objective, responsible and impartial

nature of information about human rights and humanitarian issues for whom freedom and protection should be guaranteed within the framework of national law.

Today, no state can completely block communication technologies. With the advent of satellite dishes and their cognate transmission systems and the Internet, repressive regimes can no longer murder or torture their citizens in complete secrecy. Access to sources of mass communication lifts the veil of secrecy necessary to transgress human rights. Media coverage of human rights violations is likely to prevent violations by exposing them.

News coverage of human rights shapes public opinion, affects foreign policy and serves as informal means of documenting abuse. The globalised communications system can provide human rights groups with information, assistance and support in their crusade against oppression. When people know about human rights and are aware of human rights abuses, they are more likely to seek to protect them, and exposure can lead to changes in policy by the state concerned. Human rights activists and non-governmental organisations can use global media to publicise human rights violations to shame perpetrators into changing their behaviour. This connection may also cite pamphleteering, Internet blogs, stage theatre, nukkad natak (and other traditional forms of media representation).

In this context, the case of India can also be highly illustrative with regards to human rights- media relationship. The imposition of national emergency in 1975, the atrocities committed against the Sikh community in 1984, the ongoing imbroglio in Kashmir in which the Hindu Pandits have been at the receiving end and other radical left movements symbolised in the form of Naxalism/ Maoism and other human rights abuse involving, Christians, Muslims and Dalits, starkly brings out the portrayal of human rights in media. The English language print media, in particular, has, with a few exceptions, brought out the reality in an objective manner. The vernacular print media remains much to be desired in this direction.

In all international human rights instruments, access to media has been enshrined as one of the fundamental freedoms available to all globally. Article 10 of the Universal Declaration of Human Rights (UDHR), 1948, states “everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers.” Free and open media are important factors in promoting and protecting human rights. The media can disseminate human rights information, mobilise human rights NGOs, foster popular participation in civil society, promote tolerance, and unearth government activity.

It is plausible to assert that the greater the number of communication receivers- radios, computers, newspapers, or TVs- the more informed the population. Thus, the less likely the government will be to use violence against its citizens. Suppose it is true that the media are an asset when it

comes to publicising human rights violations and possibly preventing them from happening again. In that case, it is logical to assume that the more media forms and media receivers there are in one state, the more likely the estate is to be respectful of human rights.

Among the benefits of the free flow of information and opinion are the prevention of human rights violations. The ability of the press to serve as an avenue for free expression, allowing dissenting opinions to be voiced, providing alternative information and coordinating independent explanations and analyses also benefits human rights. Controlled, intimidated, and embedded media merely promote state rhetoric and propaganda. A free press can question the state's policies. Media, under the control of national and international moneyed sections, safeguard, in reality, the interests of their respective classes. A study by a noted media specialist concludes that advances in information technology benefit human rights movements by enabling rapid transmission of information to monitor and respond to human rights violations. To advance human rights, citizens should be able to send and receive information freely. Human rights atrocities can rarely be done without public knowledge when the general public has access to TV, radio, Newspapers or Internet Communication.

Media globalisation leads to human rights globalisation. Buttressing the latter gives fillip to the notion of the universality of human rights. The UDHR, the International Covenants on Civil and Political Rights (ICCPR). 1966, to name just the two, do speak eloquently of the universality idea.

The upshot of those described above undoubtedly tells us that independent and free media plays a crucial role in maintaining human rights standards. A free press can ensure that the people will have access to information on how their government behaves towards its citizens; Independence media can reduce the secrecy needed to torture, murder, or disappear citizens by monitoring state behaviour and exposing abuses of public authority. Furthermore, media access is beneficial and necessary to maintain human rights standards. The media is difficult for the state to control, but not impossible.

2.2.2 The Media as a Tool of Human Rights Abuse

The malignant role of contemporary communication technology in encouraging human rights violations has also come to the fore in different parts of the world. The United States Institute of Peace (USIP) has, in its report of 2003, concluded that “across the globe, media have been used as tools to inflame grievances and accelerate the escalation towards violence.” The literature is galore with cases where the mass media were used to incite violence, murder and genocide, most recently in Rwanda, Somalia and Yugoslavia. They acquired international notoriety for the kind of treatment they meted out to their citizens in a not-so-distant past. With its capability of reaching large numbers of people, it can thus also be used to inflame hatred and mobilise the public to murder their neighbours. The use of the media to

inhibit human rights often requires closed or state-controlled media. Genocide, as was seen in Rwanda, necessitated state-controlled media with many receivers to incite many potential killers. A government or a military can use the media to shape an image of certain groups as enemies of the state and the people's security. Consequently, once these groups are characterised as treacherous enemies or evil predators, they can be eliminated with little impunity.

In post-independence India, the government passed the Press Council Act in 1978. Section 13 of the said Act aims to preserve the freedom of the press and maintain and improve the standards of newspapers and news agencies in India.

In most recent times, Indian media has been beset with what is rightly described as the 'crisis of content'. The crisis has manifested itself prominently in the case of the television news media, which attempts to sensationalise the issue related to women, tribals, minorities, besides other vulnerable sections of the society. The Vice President of India, While uncovering the problem, Mohd. Hamid Ansari said in July 2008: the phenomenon of convergence between news media, entertainment and telecom has meant that the demarcation between professional journalistic Output, public relations, advertising and entertainment is fast blurring. Questions are being raised whether journalists understand these demarcations and respect them or sacrifice them for commercial gain and monetary benefits. Today's public perception is that the ethical underpinning of professional journalism is weekend. The fourth estate of the Republic is at a crossroads, facing the most critical challenge in its history. The defining characteristic of the last decade has been the progressive transformation of the Indian citizen into a significant consumer of media content and products. At the broad level, one finds entertainment, especially films and reality show entertainment, increasingly thrusting itself into news media, thereby getting drawn into the culture of consumerism, a bane of western societies masquerading as globalisation. The pressure of TRP ratings has meant that astrology, superstition, crime and sleaze have prevailed, and the broad trend towards tabloidisation continues.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Define Fourth Estate? Explain its role in a democracy.

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2. Explain a key feature of media in the post- Cold war era?

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3. How do the media prevent human rights abuse?

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4. Explain the relationship between the Universal Declaration of Human Rights (UDHR) and freedom of opinion and expression?

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2.3 INTERNET

We live in a period of constant change. We can expect rapid increases in the rate of change as the population is expected to double over the next few decades. One root of this change is technology, and another source is knowledge and information. The quantum of information available to the individual is projected to double every five years. More people, more tools and more knowledge are the realities of modern life. The future promises us more changes than we have ever experienced before. The rise of information technology as an artefact of everyday life as the modern world has brought the dawn of a new era often christened as the 'Age of Information. It changes the way we perceive the world, how we think and communicates with one another, what we learn, where we work, entertain ourselves, spend free time, shop and even meet people. The Internet provides one example of a virtual environment where these changes in perceptions and relationships are constantly happening, enabling new social realities to emerge. Today, partially because we manipulate information with great ease, we are discovering the realities of our social constructs.

The Internet has become widely useful. It has been accepted much quicker with the most remarkable speed than the telephone or the radio. As the Internet forges ahead ambitiously towards critical mass, some social scientists are beginning to carefully examine the policy implication of current demographic patterns of its access and usage. Unsurprisingly, access is not distributed haphazardly but correlates strongly with income and educational levels. Looming large is the concern that the Internet may not scale economically, leading to Lloyd Morrisett, the former President of the Markle Foundation, calling a 'digital divide' between the information 'haves' and 'have-nots'.

It is assumed that most transactions in the new society will shortly be mediated by information and communication technologies. In such an environment, the ability to harness information and communication technology is critical in producing material and non-material goods and services and, more importantly, in their consumption. It also applies to social activities such as education, entertainment, art and generations of awareness.

2.3.1 Internet and Human Rights

As already mentioned, the Internet as a new medium of communication has become quite useful for disseminating information. Human Rights movements are also becoming a significant tool to generate awareness, protect international human rights standards and expose the violation of human rights both by states and non-state actors. It allows individuals to express their ideas and opinions directly to a world audience and easily to each other while allowing access to many more ideas, opinions and information than previous media have allowed. It has been observed that the Internet is a transformative force that can help open closed societies and provide the near-instantaneous flow of information to inform the public, mobilise for change and ultimately hold institutions accountable.

According to the Centre for Democracy and Technology, the effect of access to and use of global interactive media has been to promote and defend civil and political rights worldwide. Through the Internet, citizens from the most repressive regimes can find information about their government and their human rights records that newspapers may dare print.

The Government of India perceives that telecommunications and information technology is a prerequisite for the country's socio-economic transformation. It also realises that this sector has a vital role in India's Gross Domestic Product (GDP). To translate this reality into concrete action, the Government of India felt that a forward-looking telecommunication and information policy was necessary. The Government-owned media has been made autonomous through the establishment of Prasar Bharti. Now private radio and TV Channels operate without much governmental control and interference.

In 1994, The Government announced the National Telecom Policy. The Latter included, among others, the objective of Universal availability of basic telecom services to all villages. The policy contained the Internet Service Provision (ISP) by private operators who were licensed to operate and provide such services. However, the 1994 policy could not meet the declared objectives because the private sector entry into the venture has been slower than what was envisaged. An Act was promulgated on the 28th March 1997 to establish the Telecom Regulatory Authority of India (TRAI). The Department of Information Technology (DIT) has set up the National Internet Exchange of India (NIXI) in 2004 to ensure that Internet traffic originating and destined for India should be routed within India. The Latter aimed at providing an effective regulatory framework and adequate safeguards to ensure fair competition and protection of consumer interests.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. How is the Internet helpful in the promotion and protection of Human rights?

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2. Describe the negative aspects of the Internet.

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3. How can information and Communication Technology (ICT) synergies the digital environment?

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2.4 GLOBALISATION AND HUMAN RIGHTS

The international human rights movement is an integral part of globalisation. Globalisation draws people into global networks to begin to interact more regularly with others across national and cultural borders. Both globalisation and human rights claim to be universal ideas. However, each one of these universalising processes evokes resistance. Globalisation has inbuilt tendencies to shift decision-making processes away from governments and peoples to globalised economic institutions and transnational corporations with a developing interest in the socio-cultural welfare or human rights of people in the developing world. Richard Falk identifies two forms of globalisation that operate simultaneously: “globalisation from above” and “globalisation from below” While the former is imposed from the outside and is controlled and led by transnational corporations (TNCs) and international financial institutions, with states facilitating the process, the latter is local, grassroots and enjoys broader mass support as it mobilises global civil society. The two levels, according to Falk, are in a dialectical relationship. Increasingly, hegemonic neo-liberal globalisation is counterbalanced with an emerging global civil society. While globalisation from above leads to a hegemonic human rights discourse based on universalism and secularism, globalisation from below fosters a more holistic and integrated vision of rights. The latter is compatible with the worldwide realisation of human rights. While globalisation from above is inherently homogenising and hegemonic in its tendencies, globalisation from below is essentially pluralistic.

The growing activism of non-governmental organisations (NGOs) has led to domestic social and political struggles. NGOs are keen on building a culture of human rights from below, and this globalisation is a product of local agency rather than global hegemony. They are focusing on grassroots empowerment and education. NGOs have become agents of change. NGOs often construct their universalism that conforms to those advocated by

transnational networks. Many observers have emphasised that globalisation can produce either accommodation or resistance. James H Mittleman maintains that “globalisation is a multi-level set of processes with built-in structures on its power and potential, for it produces resistance against itself. In other words, globalisation creates discontents not merely as latent and undeclared resistance, but sometimes crystallised as open counter movements”. Globalisation paradoxically breeds localism and universalism as part of the same process.

Globalisation need not necessarily usher in a progressive era to promote universal human rights. Contemporary globalisation is implicated in a set of growing disjunctures between the global diffusion of the idea of universal human rights and the social, political and economic conditions necessary for their effective realisation. As the conditions of globalisation are intensified, the capacity of states and the global human rights regime to ensure compliance with the established global norms is significantly eroded. Furthermore, because globalisation creates global patterns of hierarchy and stratification, both of which accentuate growing inequalities within and between nations, it contributes to the bifurcation of the global system into zones of conformity and zones of instability. In the latter domain, internal strife, poverty, and repressive mechanisms undermine the basis of conformity with emerging global norms and hence begets chronic resistance to universal human rights standards. This domain neo-liberal vision is intimately connected to the world-historical shift in power from the nation-states to increasingly mobile types of capital and international financial institutions and international organisations. The importance of articulating democratic and progressive forms of citizenship at a national and a local level. Regional and global level as part of a wider effort to resist the profit maximising consumption-oriented, market-based citizen-subject of the neo-liberal imagination is very much a need of the hour.

The politics of human rights as distinct from politics for human rights is observable at the international level. Human rights at the latter level become the function of power dynamics. Hypocrisy is visible among the majority of the UN member states that have used the Organisation to serve their foreign policy objectives. Powerful states did not care for the UN when they perceived that it was not used to them. They took unilateral actions in the realm of international relations, impacting very gravely on the state of human rights, besides other areas.

Following the 11th September attacks in the U.S.A., countries in countering terrorism have resorted to actions that negatively impacted the state of liberties the world over. Some religious and ethnic communities have been targeted in the name of the war on terror.

2.4.1 Global Social Movements

The phenomenon of globalisation has given rise to various global social movements, one of whose objective has been the promotion and protection

of human rights. To give an institutional expression, the latter led to transnational feminist networks whose common agenda includes women's human rights, gender, justice, reproductive health rights, violence against women, peace and anti-militarism and feminist economics. These networks are variously described as global civil society entities, transnational advocacy groups, transnational social movement organisations, international and non- governmental organisations, cumulatively constituting what is described as a transnational public sphere.

Closely intertwined with those described above are what Margarted Keck and Kathryn Sikkink call 'transnational networks' or 'activists without border', that have been able to influence policies in such areas as human rights, environment and women's rights. Often led by international and domestic NGOs, they work through international campaigns, influencing the public. Government and international organisations. Keck and Sikkink trace the history of transnational development networks from the anti- slavery movements of the 18th and 19th centuries.

Globalisation is often supposed to have paved the way for the universalisation of peace since only in a tranquil world can trade and exchange of ideas flourish. But inaction, globalisation of goods, capital and people often leads to the globalisation of violence. Also, a powerful global justice movement has taken shape out of countless resistance struggles. A new transnational social activism is increasingly a potent political force in the global equation. Any attempt to understand the world and act in the world as social agents must contemplate globalisation as a concept and process. Perhaps much of the discourse on globalisation is an ideological argument for neo-liberal policies for which 'there is no alternative' (TINA). This view stresses domestic reduction in the government's economic role, deregulation, privatisation and global reduction in tariffs, opening up of capital markets, liberalisation of foreign investment regimes etc., in the context of enhanced reliance on market mechanisms. The faith market as a panacea for all society's economic ills has led critics to condemn globalisation as 'market fundamentalism'.

The Third world nation-states play an active and sometimes contradictory role in neoliberal globalisation. Usually, they are seen as the instrument of propagation for the ideology of neoliberalism. But as peripheral states, they are also constrained by imperial states. The search for how Third world states can be transformed to achieve some autonomy and political will to control the dominance of the instruments of neo-liberal globalisation-the International Monetary Fund (IMF). World Trade Organisation (WTO), General Agreement on Trade and Tariff (GATT) – has made the nation-states a contested terrain. The World Bank (WB) rhetorically claims to uphold the "satisfaction of needs" but rarely follows through in practice. In contrast, the International Monetary Fund (IMF) would say, if pressed, these needs are more likely to be satisfied by the free market. The WTO is even more caricatural since all human activities, including food, water, health

education, culture, etc., are potentially profitable commodities tradable in the world market.

By implication, state sovereignty becomes a casualty in the process of globalisation. Politically conceived, globalisation refers to the consolidation of universal institutionalisation of normative, ethical and legal governance and a cosmopolitan law of global civil society. Besides affecting other spheres of human life, the governance ideal is intimately tied to the international human rights regime, even though it contains elements of both imperial harmonisation and homogenisation. The message of globalisation is thoroughly integrated into public discourse by mass media owned by the globalists themselves, a phenomenon known as convergence.

2.4.2 Globalisation in India

In the context of India, neo-liberal economic policies have been associated with what is called reform trinity-'liberalisation, privatisation and globalisation (LPG)'. These three economic concepts have necessitated ongoing policy reforms by the Union and State governments. The architects of globalisation in India have conceived of India-centric globalisation. The content, sequence and timing of policy measures are regulated to contain potential setbacks while maximising the positive outcomes of global economic integration.

In the context of India, it is important to harness the force of globalisation to benefit human welfare and try to limit its adverse effects. Kaushik Basu, in 2001, echoed a similar view when he observed that globalisation would bring with it many ills. But on balance, it will open up more windows of opportunity for India than close. India has advantages to using globalisation as a force for development and poverty reduction. This demands a high order of Political commitment and management. Dwelling upon globalisation's social impact, Prof. Pranab Bardhan in 2001 commented that all around the world today, many advocates of social justice are in some state of despair. Some fear that social justice is a lost cause in a global economy. Thus, the LPG as conceptualised in India has been durable but somewhat distinct from its third-world counterparts.

Not with standing the above, it is pertinent to uncover the relationship between economic globalisation and its impact on human rights, particularly the poor. Those critics who view globalisation from the human rights perspective conclude that human rights have been adversely affected by globalisation. In this connection, the critics cite human development indices of the United Nations Development Programme (UNDP), independent reports by international civil society organisations such as Amnesty International, Human Rights Watch and others.

2.4.3 Globalisation Trade and Investment

The rapid increase in trade and investment over the last 20 years has offered

new hope to create growth and the resources needed to promote the full enjoyment of human rights and eradicate poverty.

Office of the High Commissioner of Human Rights (OHCHR) work on “human rights, trade and investment” has sought to analyse rules and policies on trade and investment to identify the broad areas of action at the national and international levels that would promote fairer trade that can improve the enjoyment of human rights.

The seven areas of particular relevance to human rights include:

- Agriculture
- Government procurement
- Intellectual property protection
- Investment
- Services
- Social labelling for fair trade
- Public morals and general exceptions to trade and investment rules.

Human rights principles and goals of particular relevance to trade include:

- Equality and non-discrimination
- Participation
- Accountability
- International cooperation

As a practical measure to devise trade reform that promotes the enjoyment of human rights, OHCHR has on several occasions encouraged States to undertake human rights impact assessments of trade rules and policies both during the process of trade negotiations as well as after a period of implementation of trade rules and policies. Such assessments should be public and participatory and focus particularly on the effects of trade reform on disadvantaged and vulnerable groups and the gender effects of trade rules and policies.

However, many observers point out that in practice, the development of a global economy has been matched by the development of a global society.

International institutions are not strong enough to prevent war or large scale abuse of human rights in individual countries where TNCs are operating, and they can exploit the situation for their profit motives. They can also exploit the bio-diversity and environment, running both in the long run. Global financial markets being largely beyond the control of national or international authorities, the TNCs and financial markets have grown so powerful that they can impinge upon the human rights of individuals in the society and sovereignty of the state in actual practice.

Globalisation has, thus, led to the weakening of the state and rendered the relationship between the community, the state and the corporation fluid. It has entrenched the powers and widened corporations' freedoms, eroding the powers and freedom of people in their diverse community settings. Social scientists have begun to talk of 'reinventing government' because the state is turning more one-sided in representing corporate interest and failing to represent citizen and Community interest.

For citizens and communities, the erosion of State power implies withdrawal of two protective umbrellas" to the individual citizens and the community. The first is the protection available through the regulation of (commercial) profit-seeking behaviour so that destruction of livelihoods, environment and people's health can be prevented. The second is the protection built into traditional environment rights and rights to knowledge and culture, rights which are often customary, indigenous and not written in the law, but which are central to secure livelihood and survival options, especially of marginal groups such as women, tribal, landless and small peasants, farmers, traditional healers and artisans etc.

Because of the alarming situation created by the power and reach of the TNCs, it has become imperative to re-examine the efforts of agencies and instruments promoting world trade and the global economic system. Rubens Recuperate is right in warning that if 'globalisation is to deserve its name, it has to include and not to exclude, to integrate and not to marginalise". TNCs must generate enough jobs with adequate pay and must help reduce the poverty of host and home countries. They should not rush into a heady philosophy of market profitability and economic efficiency but must bring human development and social protection into the equation. After all, the United Nations, which drew up to the Charter of Human Rights, also created World Bank and IMF to help revamp the war-ravaged economies of its member countries. The denial of these rights and fundamental freedoms is not only an individual and personal tragedy but also creates conditions of social and political unrest, sowing the seeds of violence and conflicts within and between societies and nations. In a global society, regional and local conflicts threaten to engulf the world community. But the IMF and World Bank, in their haste to create a new world economic order, failed to notice the activities of TNCs. However, their role in promoting international trade and reducing trade barriers cannot be discounted.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Explain the interrelationship between globalisation and Human rights?

.....
.....

2. Distinguish between politics of human rights and politics for human rights?

.....
.....

3. Explain the relationship between non-liberal and globalisation.

.....
.....

4. How are trade and investment relevant to human rights?

.....
.....

2.5 LET US SUM UP

In this Unit, you have read that The ‘Fourth Estate’ refers to the media. It has a right and duty, as a check on the state’s power to inform, to report truthfully about government misconduct and to expose abuses of public authority. In the post-Cold war era, the key element has been the rapidity of information flow throughout the world. States are now confronted with the glare of global public scrutiny into their human rights practices. A State that violates human rights norms, accepted by the global community in a variety of international treaties, may attract the condemnation of other countries and that of international organisations, international audiences, and its citizens. The global sharing of information can not only provide victimised groups access to supporters throughout the world, but it can also increase the costs to the estate for its repressive domestic policies towards its citizens. The media coverage of human rights violations is likely to prevent violations by exposing the violators. News coverage of human rights shapes public opinion, affects foreign policy and serves as an informal means of documenting abuse. Human rights activists and NGOs can use global media to publicise human rights violations to bring about behavioural change among the perpetrators.

We also live in a period of constant change. One root of this change is technology, and another source is knowledge and information. The rise of information technology has brought with it the draw of a new era often christened as the ‘Age of Information. Governments will face important economic, technological and political challenges to succeed in the development of internet generation. A grand challenge for the 21st century has been to access Internet facilities to all in the present context of globalisation. The Internet has become a very important tool for disseminating information and coordination for the promotion and protection of human rights.

The international human rights movement is an integral part of globalisation. Both globalisations and human rights claim to be universal ideas. Richard

Falk identifies two forms of globalisation from above” and globalisation from below”. The politics of human rights as distinct from politics for human rights is observable at the international level. Human rights at the latter level have become the function of power dynamics. Powerful states did not care for the UN when they perceived that it was not used to them.

On the other hand, the growing activism of non-government organisations (NGOs) has led to domestic social and political struggles. The neo-liberal idea stresses domestic reduction in the government’s economic role, deregulation, privatisation and global reductions on tariffs, opening up of capital markets, liberalisation of foreign investment regimes, etc., in the context of an overall enhanced reliance on market mechanisms. The faith in the market as a panacea for all society’s economic ills has led critics to condemn neoliberal globalisation as ‘market fundamentalism’. Perhaps much of the discourse on globalisation is an alibi for neo-liberal policies for which “there is no alternative” (TINA). The Third world nation-state plays an active and contradictory role in neo-liberal globalisation. It is made to propagate the ideology of neoliberalism because of TINA; dictates of imperial powers also constrain it because it is the periphery of international order.

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2.7 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

- 1) 'Fourt The Eestate' refers to the media, the other three pillars being the organs of the government, namely, the legislature, executive and the judiciary. The media have a right and duty to check the estate to report truthfully about government misconduct and expose abuses of public authority.
- 2) The key feature of media in the post-Cold war era has been the rapidity of information throughout the world. Increased information flows, rapidity of transmission and transparency are characterised, among other things, the contemporary world. Over the years, news anchors, reporters, photojournalists, and other media actors in humanitarian crises have become a common occurrence.
- 3) Media Tcho everage of human rights violations is likely to prevent violations by exposing the violators. News coverage of human rights shapes public opinion, affects foreign policy and serves as an informal means of documenting abuse. The globalisation communications system can provide human rights groups with information assistance and support in their crusade against oppression. Human rights activists and non-governmental organisations can use global media to publicise human rights violations to bring about behavioural change among the perpetrators.
- 4) Article 10 of the Universal Declaration of Human Rights (UDHR), 1948 states that everyone has the right to freedom of opinion and expression: this right includes the freedom to hold opinions without interference and seek, receive, and impart information and ideas through any media and regardless of frontiers.

Check Your Progress: 2

1. Live Wine a period of constant change. One root of this change is technology, another source of knowledge and information. This Internet has emerged as a very important tool for communication and dissemination. It can be used to have more information about Human

Rights and connect to others with similar interests. NGOs can be used for networking and fast communication with people. The Internet provides an environment where these changes constantly occur, enabling new social realities to emerge.

2. Roris Tts and fundamentalists can use the Internet to propagate hate and violence by exploiters to use children as sex objects and vested interest to practice discrimination.
3. Information and Communication Technology (ICT), one among many tools, helps secure broader and lasting availability of world information resources, which the industrial society has grossly overlooked. Its transmission rules don't follow consumerism, but it might greatly enhance the digital environment if it adopted those rules. To fully exploit ICT for creating local and relevant content, people need far wider skills than simple web page creation. To sustain an information society, adequate attention needs to be geared towards intellectual and human requirements, capabilities and values. Learning how to apply ICT is necessary to better one's life. In a few decades, there have been advances in the general application of ICT, and some societies have developed a notably superior ICT infrastructure. A basic ICT infrastructure is one of the requisites for interaction, though it does not necessarily mean that each adult should be equipped with all the devices in existence. But access to and use of ICT should be autonomous, voluntary and natural in the first place. The idea of building an overall knowledge-sharing instrument at the initiative of a single authority should be avoided.

Check Your Progress: 3

1. Intern Tathioenal human rights movement is an integral part of globalisation. Globalisation draws people more closely into global networks than ever before. Both globalisation and human rights claim to be universal ideas. However, each one of these universalising processes evokes resistance. Globalisation has inbuilt tendencies to shift decision-making processes away from governments and peoples to the globalisation of economic institutions and transnational corporations that have only a negligible interest in the human rights of people in the developing world. Richard Falk identifies two forms of globalisation that operate simultaneously: "globalisation from above" and "globalisation from below". While the former is imposed from the outside and is controlled and led by transnational corporations (TNCs) and international financial institutions, with states facilitating the process, the latter is local at grassroots and enjoys broader mass support as it mobilises global civil society. While globalisation from above is inherently homogenising and hegemonic in its tendencies, globalisation from below is essentially pluralistic.
2. Politic shoe of human rights as distinct from politics for human rights is observable at the international level. Human rights at the latter level

have become the function of power dynamics. Powerful states did not care for the UN when they perceived that it was not used to them. They took unilateral actions in the realm of international relations, impacting very gravely on the state of human rights, besides other areas. On the other hand, the growing activism of non- governmental organisations (NGOs) has led to domestic social and political struggles. NGOs are keen on building a culture of human rights from below, and this globalisation is a product of local agency rather than global hegemony. Focusing on grassroots empowerment and education, NGOs have become agents of change. NGOs often construct their universalism that conforms to those advocated by transnational networks.

3. Neo-liberal idea stresses domestic reduction in the government's economic role, deregulation, privatisation and global reductions on tariffs, opening up of capital markets, liberalisation of foreign investment regimes, etc., in the context of an overall enhanced reliance on market mechanisms. The faith in the market as a panacea for all society's economic ills has led critics to condemn neoliberal globalisation as 'market fundamentalism'. In its neo-liberal global garb, global capitalism results in ever-increasing inequality, immiseration and alienation- 'globalisation's discontents. The instruments of neoliberal globalisation. The International Monetary Fund (IMF), World Trade Organisation (WTO) and United Nations Conference on Trade and Development (UNCTAD) have ensured its peripheral role.
4. Human rights, trade and investment, has sought to analyse rules and policies to promote fairer trade that improves the enjoyment of human rights. The seven areas of particular relevance to human rights include:
 - a) Agriculture
 - b) Government procurement
 - c) Intellectual property protection
 - d) Investment
 - e) Services
 - f) Social labelling for fair trade
 - g) Public morals and general exceptions to trade and investment rules.

UNIT 3 ICTs AND WOMEN (ISSUES OF ACCESS AND EQUITY)

Structure

- 3.0 Introduction
- 3.1 Learning Outcomes
- 3.2 Gender Issues in ICT
- 3.3 Women's Access to ICTs
 - 3.3.1 Strategies for Gender Equity
- 3.4 Benefits of ICTs for Women
- 3.5 Let Us Sum Up
- 3.6 Keywords
- 3.7 Further Readings
- 3.8 Check Your Progress: Possible Answers

3.0 INTRODUCTION

By definition, Information and Communication Technologies (ICTs) are a diverse set of technological tools and resources to create, disseminate, store, bring value- addition, and manage information. Traditional ICT tools such as television, radio and the telephone have proved their effectiveness in promoting development in marginalised areas. The emergence of computers, the Internet and wireless communications technology, and powerful software for processing and integrating text, sound and video into electronic media comprise modern ICTs. The spread of the global electronic network of computers popularly referred to as the Internet and mobile connections, have generated an unprecedented global flow of information, products, people, capital and ideas. It is the fastest-growing industry globally, and it is poised to become the largest in the world. The ICT revolution impacts economic and social conditions around the world, cutting across geographical locations and state of development of countries around the globe. ICTs are also proving to be a vital tool in helping to link new civil society networks around key issues, from global warming to women's empowerment.

Despite the vast opportunities for economic growth and social development offered by ICT, there is a negative aspect. ICT may further widen the gap between developed and developing countries, between the rich and the poor and between those who know how to use the new technologies and those who do not. The digital divide in access to ICT between the developed and developing world results from various factors, including poverty, lack of resources, illiteracy and low levels of education. In many societies, women are the most deprived, with the least access to resources and little control over

decisions affecting their lives. For this reason, women are on the wrong side of the digital divide, with limited access to and control over ICTs. This concern has led the United Nations and other development organisations worldwide to work cooperatively with developing countries to build “digital opportunities” to foster sustainable human development and reduce poverty.

The focus of the present Unit is to examine women’s issues in accessing and making use of ICTs for their advancement in society as a whole.

3.1 LEARNING OUTCOMES

The world is in the midst of a knowledge revolution. It witnesses the opening up of entirely new vistas in communication technologies, which are indeed revolutionary in nature. ICTs have the potential to enhance capabilities and create a new types of economic activities and employment opportunities and thus enhance the quality of life. Besides that, it leads to enhanced participation of citizens in diverse areas of social life. However, society’s technological, economic, and social development is possible through the skilful use of ICTs. How women are placed in terms of equity and access to technologies in ICTs lead knowledge society is the focus of this Unit.

After reading this Unit, you will be able to understand:

- the gender divide within ICTs and its implications;
- the gender disparity which exists in equity and access to ICT; and
- the potential benefits of ICTs for women and the strategies for achieving it.

3.2 GENDER ISSUES IN ICT

ICT is for everyone, and women have to be equal beneficiaries of the advantages offered by the technology, products and processes, which emerge from their use. Undoubtedly, ICTs have opened up new opportunities for women to communicate and share information. The services provided by ICTs have allowed women greater access to information for their productive and reproductive tasks and for organising and expanding their local organisations, networking and linkage building, locally and internationally.

While the potential of ICTs for stimulating economic growth and socio-economic development, and effective governance is well recognised, the benefits of ICTs have been unevenly distributed within and between countries. This uneven distribution is termed as ‘digital divide’. It refers to the differences in resources and capabilities to access and effectively utilise ICTs for development that exists within and between countries, regions, sectors and socio-economic categories.

While there is a recognition of the potential of ICT as a tool for the promotion of gender equality and the empowerment of women, grave

inequality exists in terms of accessibility of ICTs based on gender identity. An insufficient number of women access and use ICT compared to men. Initially, the gender divide was identified in 1995 by the United Nations Commission on Science and Technology for Development (UNCSTD). The Commission identified significant gender differences in access levels to, control of and advantages accruing from a wide range of technological developments.

Analysis of gender issues in ICTs builds on previous gender analysis of technology. History proved that technological developments are not value-neutral. They are socially constructed and thus have differential impacts on men and women.

Women's capacity to exploit the potential of the new ICT as tools for empowerment is constrained in different ways. Some of these constraints are linked to factors (such as technical infrastructure, connection costs, computer literacy, language skills etc.) that affect men and women alike. Still, due to the vast existence of discrimination that persists in society, the disadvantages are exacerbated for women. Thus besides technological infrastructure, the socially constructed gender roles and relationships play a crucial role in determining the capacity of women and men to have access to ICTs on equal terms.

Do You Know?

Advocacy for gender issues in information and communications technologies first gained a foothold during the Fourth World Conference on Women in Beijing in 1995. Women's organisations successfully lobbied to recognise the need for women to be involved in decision-making regarding the development of new technologies to participate fully in their growth and impact. The recommendations in the Platform focused on increased access and participation of women to expression and decision-making in the media and information and communication technologies to overcome negative portrayals and stereotypes of women in media and communications.

www.wigsat.org/malcolm.html

Economic and other market factors also play a crucial role. Historically, the isolation of women from the mainstream economy and their lack of access to information because of societal, cultural and market constraints have led them to become distant from the global pool of information and knowledge. This distance is reflected in the levels of empowerment and equality of women compared to men. Thus the capabilities of women to effectively use ICTs for their advancement is dependent on many social, economic and structural factors, including education, literacy, geographic location, social class etc.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

- 1) Discuss the progress and challenges in transforming gender relations in ICT dominated contemporary society.

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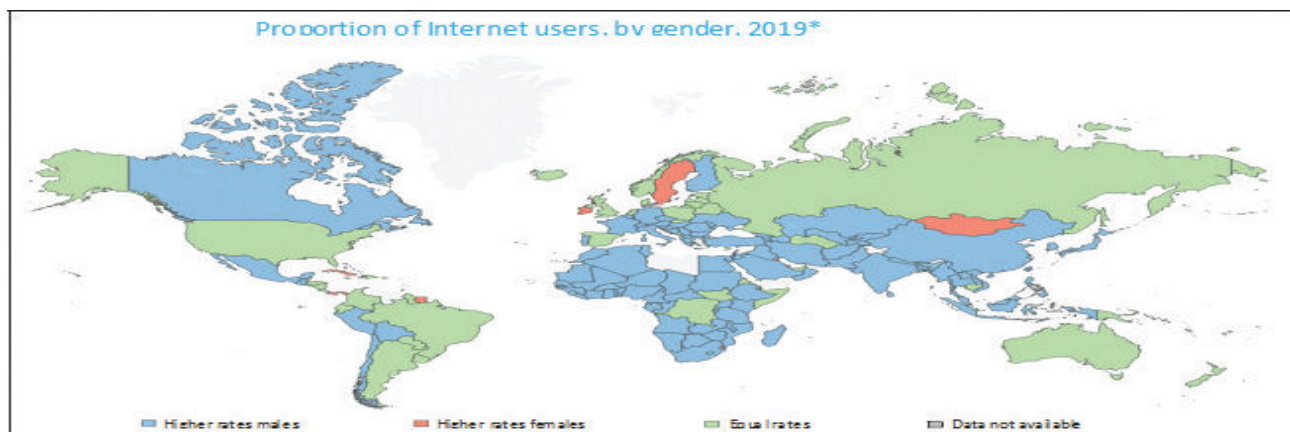
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3.3 WOMEN'S ACCESS TO ICT

From the discussion in the preceding section, we could gather that access to ICT is typically divided along traditional lines of unequal distribution of benefits of development, resulting in unequal access that has become known as the 'digital divide' or 'digital exclusion'. The digital divide reflects both old socio-economic and political divisions exacerbated by the introduction of new ICT and new divisions created due to the nature of the latest ICT. Women are at the deepest end of the digital divide has been the central message of gender advocates working in the field of ICT.

It is a hard truth that most of the poor are women, and they experience vulnerability and powerlessness to a much higher degree than men. Therefore, equitable access to ICT technology and the autonomy to receive and produce the information relevant to their concerns and perspectives are critical issues for women. Women's access to ICTs and control of them (or lack thereof) is dependent on many factors. Factors such as gender discrimination in jobs and education, social class, illiteracy and geographic location (urban or rural) influence women's access to ICTs or any other sort of modern communication system. Let us see some of the main constraints women have regarding accessibility to ICTs.

The sophistication of any ICT infrastructure introduced into any environment becomes meaningless if women do not have the skills to operate the system and use it to their best advantage.



Source: <https://itu.foleon.com/itu/measuring-digital-development/gender-gap/>

Infrastructural Constraints: The potential of ICTs for women depends on their levels of technical skill and education and is the principal requirement for accessing knowledge. The Internet, the fastest-growing category of contemporary ICTs, is linked to literacy. Internet technology is foreign, with English being the predominant language, which becomes a handicap.

There is no access to more and higher education without basic literacy, much less to ICTs. The challenge of illiteracy must be overcome before women can benefit from ICTs. Women have less time to learn because of heavy domestic chores and other socio-cultural factors. Increased female enrolment at the primary level will enhance female participation in higher education, which enhances female participation in technological education. As a result, more women could be benefited from the new avenues created by ICTs and hence become empowered.

Necessary interventions are carried out, especially by the government, and the NGOs can lead to skill development and rising education levels among women. It could be done by imparting technical education on the use of ICT as part of both formal and informal educational systems and initiating distant-learning and vocational courses.

Locational Disadvantages: There is a definite urban bias in ICT access and use by women. Access to IT education and training is highly skewed towards urban-centred, English-speaking masses with a comparatively higher socio-economic status. Infrastructure is concentrated in urban areas. If technologies are made that have an urban bias and high cost, few women would have access. Better Internet connectivity is available within urban areas, while most women reside in rural areas. Simply by being in the majority in rural areas, women have a small chance of accessing new technologies.

[For a detailed and generic discussion on Internet access and digital inequality, refer to Unit 3 of MNM011 - Issues and Participation of ICT]

Economic Constraints: The new technology comes at a financial cost which hinders its penetration to the individual and sometimes even at the community level. The problem is even compounded because poverty and lack of economic power are borne more by women than men. They have little control over the household income and do not have the decision-making power to invest in these technologies. Women are less likely than men to own radios and TV or access them when they want. Access to the Internet is through an Internet Service Provider, or an Internet café is expensive. Further, there are associated physical and infrastructural requirements such as electricity, mobile ownership, and quality of 2G/4G connectivity, which are unevenly distributed and add to the cost of initiating knowledge networking.

Structural Constraints: Women's access will be limited if the information centres are located in places that women may not be comfortable frequenting. Specifically, access for rural women will depend critically on where the technologies are located. To facilitate access for women from various classes and sectors, ICTs will need to be located in places that women frequent and

have open and equal access, such as health centres, women's NGOs, women's employment centres, and perhaps even places of worship. In this context, location also pertains to the practical, specific kind of information that women require due to the time constraints they face. Establishing telecommunication centres in local communities is also a potentially useful strategy if gender obstacles to access to them by women are removed into account. Information production and distribution strategies will also be an important consideration to make the most of each point of access. They will need to be flexible, mixed media and multi-technology systems to effectively reach the greatest number of women.

Social and Cultural Factors: Many women continue to be intimidated by the Internet and its technology, finding it more of an area best left to the men. This stems from social conditioning and the fact that many young women are not encouraged to take up science subjects in school (especially in rural areas) or feel that it would be an area in which they could not excel. This is an aspect of gender socialisation. Similarly, limited awareness of the full range of opportunities afforded by ICT other than simple passive access to information; lack of understanding as to how ICT can be used actively to disseminate data, lobby, participate in and influence decision-making processes, coordinate community activities and collaborate with other NGOs at local and regional levels are often cited as barriers facing women.

3.3.1 Strategies for Gender Equity

Think It Over

What are the constraints that women have regarding the accessibility of modern information and communication technologies?

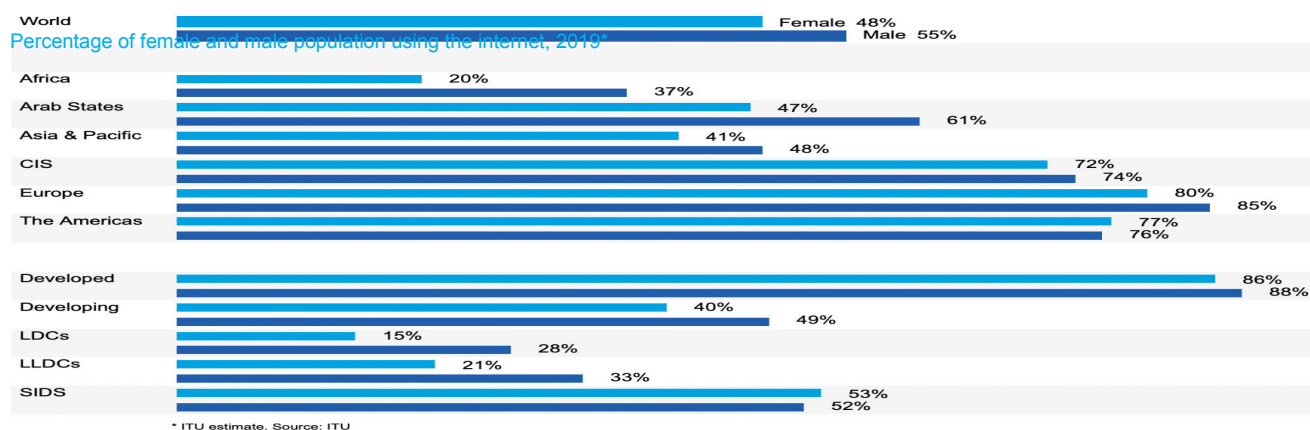
To ensure gender equity with regard to access to and use of ICTs by women and men, different areas need to be addressed. Interventions have to take place simultaneously at different levels. Development agencies should assist national policymakers and implementers in developing countries in elaborating and implementing gender-sensitive ICT policies, strategies, and activities through policy advice, capacity building, and financial support. (Hafkin 2001)

- **Creating a conducive policy and an enabling environment:** Though the need for policies on information and communication technologies have been realised, very few policies promote gender equity in using ICT. Gender concerns have to be considered in policy content and the process of policy development.
- **Improving women's access to ICT:** To ensure women's access to ICT, there is a need to increase the availability of communication in presently underserved areas. Since women concentrate in rural or semi-urban areas. Emphasis has to be put on common use facilities. The most sustainable strategy is a market-driven, locally initiated service

model. Internet services could be added to existing communication services of phone shops, teleports, public call offices etc. Public access to communication facilities should be provided in or near institutions that women frequent, such as schools, markets, health clinics or post offices. Mobile units could also be used to provide communication services to underserved communities.

- Improving girls' and women's access to education and training:** The most important factor in increasing the participation of women and girls in ICT is to offer them more education at all levels, from literacy to scientific and technological education. Women and girls need to be prepared for various roles in ICT as users, creators, designers and managers. Efforts have to be made to increase the number of girls and women studying ICT-related subjects in formal and outside schools. This includes reviewing training methodology and contents. The possibility of scholarships helps to encourage women to enter science and technology-based subjects. Women outside the formal schooling system and already part of the workforce should be given access to upgrading and retraining courses or the possibility to acquire new skills. Training providers should be encouraged to address a female clientele and develop appropriate training programmes specifically.
- Providing access to capital:** Linking entrepreneurs to ICT requires education and training and the provision of finance. In this context, microfinance schemes are the most likely sources of capital. Managers of microfinance institutions need to be made aware of the viability of ICT-based businesses of women. The example of Grameen Telecom's Village Phone Programme in Bangladesh illustrates how even the poorest, including women, can successfully set up ICT-based businesses.

Internet gender gap large in developing countries



Think It Over

How can the current uneven and unaffordable access to information by women be improved through ICT?

Regarding the Indian situation, the following recommendations relate to strategy and lines of action that will enable women to overcome the many obstacles they face and help guarantee them more equitable access to new and emerging communications technologies and electronic information sources (www.egovindia.org).

- Promote the access of women, girls and women's organisations to new and emerging communications technologies and computerised information resources;
- Promote the development of computerised information resources on issues related to the advancement of women;
- Support the development of initiatives of women and citizens' groups in the field of computer networks that promote the advancement of women and gender equality;
- Support women and girls' access to training in using computer networks and promote a gender perspective in training and methodology in the field of new technologies;
- Promote equal access of women to advanced technical training and careers in computer communications;
- Promote and support the equal participation of women in international and national decision-making relating to the use of communications infrastructure and access to computer networks;
- Create content that reflects women's needs and voices; and
- Facilitate and encourage the involvement of women in technological innovation.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Discuss what role NGOs, civil society organisations, and other networks play in creating equality for women and disseminating gender-related information.

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3.4 BENEFITS OF ICTS FOR WOMEN

Women can benefit from ICT in myriad ways. Let us try to list some of them here.

- **Influencing public opinion on gender equality:** Through ICT enabled information channels including radio, telephone, television and the Internet, awareness of gender equality issues can be raised and enhanced.
- **Increasing economic opportunities for women:** Women have been limited from participation in many forms of economic life. ICTs help entrepreneurs reduce transaction costs and increase market coverage.
- **ICTs for women's education:** ICTs can provide women with useful information. Radio, television broadcasting, audio and video cassettes are used as supplements to print materials in traditional education. But now newer technologies such as audio and video conferencing, the Internet and e-mail are taking over.
- **Encouraging women to know their rights and participate in decision-making:** Women remain vastly under-represented in national and local assemblies. ICTs have the potential to empower women by enabling them to participate in public discussions.

Engendering knowledge networks opens up avenues for women to freely articulate and share their experiences, concerns and knowledge with the possibilities of their further enrichment as the same pass through a gamut of network users. They are instrumental in helping women break from society's stereotypical structures, narrow outlooks, and the hegemony of male-dominated societal structures. Other benefits include objective and targeted information flows, low communication costs, sharing best practices and solutions, opening up alternate communication channels with women, hitherto unreached or under-served, and accomplishing a deeper geographic penetration.

Through the use of convergence and hybrid technologies such as community e-mails, community radio broadcasts, telecentres, newsletters, videos, etc., women communities could overcome the constraints of seclusion, mobilise resources and support, reach out to new markets, and open up avenues for life-long learning.

Do You Know?

Gyandoot is an Internet project in the Dhar district of Madhya Pradesh, connecting 21 rural cyber cafés called Soochanalayas. Women can send complaints with the assurance of reply within a maximum period of seven days (www.gyandoot.net). Asian Women Resource Exchange (AWORC) is an Internet-based women's information network and service in Asia. AWORC develops cooperative approaches and partnerships to increase

access to and explore ICT applications for women's empowerment ([HTTP:// www.aworg.org](http://www.aworg.org)).

Andhra Pradesh State Wide Area Network (APSWAN) is the backbone for the voice of women, data and video communication throughout the state of Andhra Pradesh. This network connects the state secretariat with 25 centres, including all districts headquarters town through e-mail and video conferencing. Women stand to enormously benefit from them as they now have the power to write directly to the Chief Minister, cross-cutting the hierarchical layers (Source: www.ap-it.com/apswan.html)

ICTs open up a direct window for women to the outside world. Information now flows to them without distortion or any form of censoring, and they have access to the same information as to their male counterpart. This leads to broadening perspectives, building up a greater understanding of their current situation and causes of poverty, and initiating interactive processes for information exchange. Further, such forms of networking open up alternate forms of communication to those offered by the conventional or the government-controlled media sources and therefore catalyses the empowerment process. The opening up of alternate forms of communication with the external world made the women more informed. They were empowered to realise that their real causes of poverty were not natural disasters but ineffective state governance mechanisms. Therefore, a link was established between bad governance and poverty - their first step to empowerment, as they were able to identify the causal loop to their poverty and the players involved.

New areas of employment such as telemarketing, medical transcription etc., have also opened up tremendous job opportunities for women. These jobs are underpaid and fall at the lower segment of ICT jobs; nevertheless, they open up avenues where none existed before. One of the most powerful applications of ICT in knowledge networking is electronic commerce. Electronic commerce refers to the selling of products and services online and the promotion of a new class of ICT-savvy women entrepreneurs in rural and urban areas. Women over time have learnt the advantages offered by ICT and its potential in opening up windows to the outside world. This has put them in greater control over the activities performed by the laying the foundation of entrepreneurship development.

The unrestricted flow of information through ICT processes opens up avenues for men and women to view each other differently. The sharing of views between communities living in different geographical and cultural spheres will lead to broadening views and changing mindsets over time. It is a fact that the horizontal level of communication has a greater impact than the vertical communication structures, and knowledge networking promotes the horizontal flow of information. Men may learn more about women's productive roles in the wider economy in different cultures and regions and

may become more willing to provide equal spaces to women. Removing this stereotypical mindset would undoubtedly be a big step towards women's empowerment.

Do You Know?

India's Information Villages at Pondicherry

Popularly known and showcased as best practice in ICT, this project was initiated by the M.S. Swaminathan Research Foundation (MSSRF) and NGOs in Pondicherry State in South India. It was initially undertaken for poor agricultural and fishing villages in South India. It involved the setting up information centres, with telephone facilities and internet services, in 9 villages in Pondicherry. The Information centres have databases from which women can access information on agricultural concerns, general hygiene, employment, and income-generating opportunities. At present, their databases, which women can access through their information centres, now include Entitlements to Rural Families, Grain Prices in the Pondicherry region, Information on Seeds and Fertilisers, Insurance Schemes, Integrated Pest Management in Rice Crops and Sugarcane crops, Directories of Hospitals and Medical Practitioners in Pondicherry grouped by specialisation; Bus/ train timetables covering the Pondicherry region and two nearby towns.

The success is attributed to its participatory and gender-sensitive processes, the transfer of technical skills to the rural community, and the close partnership of the NGOs in Pondicherry and the M.S. Swaminathan Research Foundation.

Source: Jimenez-Tan, Marion (October 2001), "Appropriating ICT For and By Women in Asia and the Pacific" APGEN-UNDP Paper Presentation for the Asia Pacific Regional Workshop on Equal Access of Women in ICT, Seoul, ROK

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. How can ICT enhance women's socio-economic and political development participation, leading to a knowledge society?

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3.5 LET US SUM UP

Despite advances in the ICT industry, the benefits of ICT have not reached most people. To a large extent, the traditional pattern of male and female attitudes towards technologies replicates itself in developing the new ICTs.

ICT cannot be effectively used for the development of society as a whole and women especially, unless the crucial problem of the digital divide is addressed, not only between developed and developing countries but also in terms of gender, class, age, ethnicity, language, geographical location and physical ability. Women and girls must be explicitly included as beneficiaries of the 'ICT revolution' as a fundamental principle of equality and an essential element in the shaping, direction and growth of the 'Information Society. They must have equal opportunities to actively participate in ICT policy decision-making spaces and the agenda-setting processes, which shape them.

The focus of this Unit is on ICT and women. It discusses the main issues that concern women in the ICT based present-day development of society. It further discusses some of the strategies to involve women in the kind of development that takes place in contemporary society. One of the main strategies is to have a gender perspective in the national ICT policies.

How ICT can support gender empowerment is also discussed in this Unit. Suppose access to and use the new technologies is directly linked to social and economic development. In that case, it is essential to ensure that women understand the significance of these technologies and use them. If not, lack of access to ICT becomes a significant factor in their further marginalisation from the economic, social and political mainstream, both within their countries and the world. Without equal and full participation in ICT use, women will be left out from participating in the global world of the 21st century.

3.6 KEYWORDS

- Civil Society** : Civil society or civil institutions refers to the totality of voluntary civic and social organisations or institutions that form the basis of a functioning society instead of the force-backed structures of a state. Examples of groups in civil society include universities, non-governmental organisations, environmental movements, indigenous peoples' associations, organised local communities and trade unions. Civil society can be organised at the local, national and international levels.
- Internet Café** : It is a place where one can use a computer with Internet access free of cost or payment.
- Information Society** : It is a society in which the creation, distribution and manipulation of information or knowledge is a significant economic and cultural activity.

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3.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Analysis of gender issues in ICTs builds on previous gender analysis of technology. History proved that technological developments are not value-neutral. They are socially constructed and thus have differential impacts on men and women. Women's capacity to exploit the potential of the new ICT as tools for empowerment is constrained in different ways. Some of these constraints are linked to factors (such as technical infrastructure, connection costs, computer literacy, language skills etc.) that affect men and women alike. Still, due to the vast existence of discrimination that persists in society, the disadvantages are exacerbated for women. Thus besides technological infrastructure, the socially constructed gender roles and relationships play a crucial role in determining the capacity of women and men to have access to ICTs on equal terms.

Check Your Progress: 2

- 1 To ensure gender equity with regard to access to and use of ICTs by women and men, different areas need to be addressed. Interventions have to take place simultaneously at different levels. Development agencies should assist national policymakers and implementers in developing countries in elaborating and implementing gender- sensitive ICT policies, strategies, and activities through policy advice, capacity building, and financial support.

Check Your Progress: 3

1. Engendering knowledge networks opens up avenues for women to freely articulate and share their experiences, concerns and knowledge with the possibilities of their further enrichment as the same pass through a gamut of network users. They are instrumental in helping women break from society's stereotypical structures, narrow outlooks,

and the hegemony of male-dominated societal structures. Other benefits include objective and targeted information flows, low communication costs, sharing best practices and solutions, opening up alternate communication channels with women, hitherto unreached or under-serviced, and accomplishing a deeper geographic penetration.

**ICT's and Women
(issues of Access
and Equity)**



UNIT 4 OPEN SOURCE MOVEMENT

Structure

- 4.0 Introduction
- 4.1 Learning Outcomes
- 4.2 History of Open Source
- 4.3 Open Source Movement
 - 4.3.1 Open Source Software
 - 4.3.2 Open Source Software: Development Path
 - 4.3.3 Open Source Software vs Commercial Software
- 4.4 Open Source Software: Philosophy, Principles and Licensing
 - 4.4.1 Philosophy of Open Source Software
 - 4.4.2 Open Source Software (OSS)
 - 4.4.3 Licensing of Open Source Software
 - 4.4.4 Open Source and Open Standards
- 4.5 Types of Software
- 4.6 Desirable Software Attributes
- 4.7 Advantages of Open Source Software
- 4.8 Legal Issues
 - 4.8.1 Copyright
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- 4.9 Other Successful Open Source Software
- 4.10 Applications of Open Source in Other Fields
- 4.11 Let Us Sum Up
- 4.12 Further Readings
- 4.13 Check Your Progress: Possible Answers

4.0 INTRODUCTION

Open source is a set of principles and practices that promote access to the design and production of goods and knowledge. The term is commonly applied to source code knowledge. The term is commonly applied to the software's source code available to the general public with relaxed or non-existent intellectual property rights. This allows users to create software content through incremental individual effort or collaboration. Open source denotes that the origins of a product are publicly accessible in part or whole. The term open source refers to software in which the source code is freely available for others to view, amend and adapt. Typically it is created and maintained by a team of developers that crosses international and national boundaries.

Technically speaking, the software is made of lines of instructions to computers called ‘code’. Depending on the complexity of the software, its code can be a few dozen lines or several million lines. This code is written following the rules of a certain grammar, called a language like C++ or Java or Python. Changing or modifying the code can extend or modify the software’s features. Commercial companies like Microsoft and Oracle view the code as their intellectual property and protect it to the extent of not allowing modification even by their customers who have paid to buy and use it. Like those behind Linux Operating System or OpenOffice, open- source groups consider software as a ‘building block’ that can and should be modified by its users to customise it for their particular use.

4.1 LEARNING OUTCOMES

After studying this Unit, you should be able to:

- explain the term open-source;
- describe the difference between open-source software and proprietary software;
- list the advantages of open source software;
- describe legal issues which pertain to such software; and
- know the application of open source in other fields.

4.2 HISTORY OF OPEN SOURCE

The history of open source is a very interesting one. To start with, in the beginning, there was only one free (Libre) software. Later on, proprietary software was born, and it quickly dominated the software landscape to the extent that now it is usually considered the only possible model by many people. However, in recent times the free software has again gained popularity.

During the 1960s, when IBM and other companies were selling large scale commercial computers, they came with some free software (Libre), free in the sense that it could be freely shared among users. This came with a source code, which could be improved and modified. However, in the late 1960s and mid-’70s, the proprietary software started making its place in the market.

In the late 70s and early 80’s Richard Stallman, an American software developer who had an idea that sharing source code and ideas are fundamental to freedom of speech developed a ‘free’ version of the widely used ‘UNIX’ operating system. This resulted in a ‘GNU’ program released under a specially created General Public Licence (‘GNU.GPL’). The GPL is the license under which much Open Source code is distributed. It enshrines the idea that computer code, like speech, should be free to be copied, interpreted, modified and generally mashed up. This was designed so that source code could remain openly available to all. This software was designed for commercial usage or distribution. Such an approach was called ‘free software. Thus the term “free” denotes that anyone could modify the software as per his/her needs.

During the 80's and the early 90's open-source software continued its development. USENET and the Internet helped transnational coordinate efforts and built strong user communities.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Explain the basic philosophy behind the open-source concept?

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4.3 OPEN SOURCE MOVEMENT

This section systematically covers the definition, scope, and origin of open source software, including the fundamental differences between open source and closed source software.

4.3.1 Open Source Software

Open Source Software (OSS) is not a new idea. You already know that the open-source movement started with the Internet. Recently, technical and market forces joined together to draw a niche role of the open-source movement. The open-source movement has all the potential to define the next century's computing infrastructure (Marco & Lister, 1987). Open source is a software development model and a software distribution model. OSS development follows Linus Torvalds's (Linus Torvalds is the developer of Linux operating system – an open-source system software) style of development – *release early and often, delegate everything and be open to the point of promiscuity*. Raymond (2001a; 2001b) termed this type of software development as a bazaar-style of development in comparison with the traditional software development process (termed by Raymond as cathedral model), which is carefully crafted by individual wizards or small group of experts working in splendid isolation. The Open Source Initiative (2004), a forum to promote open-source software movement as a viable alternative to commercial software claims –

“This rapid evolutionary process produces better software than the traditional closed model, in which only a very few programmers can see the source and everybody else must blindly use an opaque block of bits.”

OSS is also considerably different from shareware, public-domain software, Freeware, or software viewers and readers made freely available without access to source code. Whether or not one registers it and pays the registration fee, shareware typically provides no access to the source code. Unlike freeware and public domain software, OSS is copyrighted and distributed with license terms designed to ensure that the source code is

always available. Sometimes, a small fee may be charged for the software's packaging, distribution, or support.

Definition

The open-source movement has been in conscious development for nearly two decades, but the term “open source” itself has been a relative latecomer. Christine Peterson of the Foresight Institute proposed the term *open source* in late 1997 during a meeting of a small group of open-source movement key persons (Raymond, 2001c). This group registered the domain name *opensource.org*, defined “open source,” developed Open Source Initiative (OSI) group, designed OSI certification, and created a list of licenses that meet the standards for open source certification. In the open-source software development model, the software's source code is made freely available along with the binary version so that anyone can see, change, and distribute it subject to the condition he/she abides by the accompanying license. According to OSI (Open Source Initiative, 2003a) –

“Open source promotes software reliability and quality by supporting independent peer review and rapid evaluation of source code. To be certified as open source, the license of a program must guarantee the right to read, redistribute, modify, and use it freely”.

Analysis of definitions given by Chudnov (1999), Raymond (1996), Moody (2001), and Morgan (2002) identifies the following attributes of OSS –

- OSS is typically created and maintained by developers crossing institutional and national boundaries, collaborating by using Internet-based communications and development tools;
- OSS development process follows the famous Linus's law – “Release early, release often and listen to users”;
- Quality, not profit, drives open source developers who take personal pride in seeing their working solutions adopted; and
- Intellectual property rights to open source software belong to anyone who helps to build it or simply use it and is not locked to any single vendor or institution.

4.3.2 Open Source Software: Development Path

The computing community started realising the advantages of sharing source codes in the late 1970s by using the Internet as a platform. The early 1980s witnessed a big conflict between OSS and proprietary software. For example, MIT Artificial Intelligence Lab established Symbolics in the early 1980s and made all the freely available software proprietary under its name. This conversion process eventually killed the culture of code- sharing at MIT Lab. This destruction is important in OSS history because it initiated the free software movement through the formation of the Free Software Foundation (FSF). Richard Stallman, one of the MIT lab members, started The GNU (recursive acronym for GNU is Not Unix) project (a free operating system) in

January 1984 and established FSF in 1985 to promote Free Software and the GNU project. The next big contribution to the free software movement came from a student in 1991. Linus Torvalds, who was a second-year graduate student at the University of Helsinki, wrote a Unix- like kernel (Kernel is a core part of the operating system) and named it Linux. He distributed Linux widely, considered users as co-developers and improved it considerably quicker. Linux kernel soon adapted to become the core of the GNU/Linux operating system, and many other parallel projects (like BIND, Perl etc.) merged with it. In 1997 GNU/Linux became the buzzword in the computing community because, within five years, it owned 25 per cent of the server market and grew at the rate of 25 per cent per annum. It's now clear that the code sharing and free software culture has been in conscious development for nearly three decades since the beginning of the Internet. But the term "open source" has been a relative latecomer. Christine Peterson of the Foresight Institute proposed the term *open source* in late 1997 during a meeting of a small group of open-source movement key persons (Raymond, 2001a). This group registered the domain name *opensource.org*, defined "open source," developed Open Source Initiative (OSI) group, designed OSI certification, and created a list of licenses that meet the standards for open source certification.

4.3.3 Open Source Software vs Commercial Software

The whole array of software can be grouped into two fundamental categories—system software and application software. System software (such as operating systems) is responsible for the overall management of computer resources. In contrast, application software is designed to perform certain tasks and thereby make computers able to perform different predefined jobs. This division is based on the application domain of software. The distribution policy may be grouped into two broad divisions – close source software and open-source software (OSS). Open-source software is also known as Free/Open Source Software (FOSS) or Free/Libre Open Source Software (FLOSS). Closed source software may again be placed in two groups – commercial software and Freeware. So, as per the distribution policy (as mentioned in the beginning), the whole array of software may be categorised into three groups – Commercial software, Freeware, and Open- source software.

Commercial software	Freeware	Open-source software
Only binary code is available against fees	Only binary code is available at no cost	Both source code and binary codes are available at no cost
As source code is not available, customisation is not possible	As source code is not available, customisation is not possible	As source code is available, extensive customisation is possible and allowed

The license agreement allows only the use of software for a definite period, and it is mandatory	The license agreement allows to use for an indefinite period, and it is optional	The license agreement allows to use, change, modify and distribution of software for an indefinite period, and it is mandatory
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4.4 OPEN SOURCE SOFTWARE: PHILOSOPHY, PRINCIPLES AND LICENSING

You already know from the previous section what open source is and how it differs from other software distributions, including a brief history of the open-source movement. In this section, we will study philosophies and principles of open-source software, IPR issues related to open source and the application of open standards in open source software development.

4.4.1 Philosophy of Open Source Software

The open-source software world is dominated by two major philosophies: the Free Software Foundation (FSF) philosophy and the Open Source Initiative (OSI) philosophy. The philosophy of FSF centres around four user-driven freedoms –

- the freedom to run a program, for any lawful purpose;
- the freedom to study how a program works and adjust it to specific needs (obviously, access to the source code is a precondition for this);
- the freedom to redistribute software; and
- the freedom to improve a program and distribute a modified program (again, access to the source code is a prerequisite for this).

Therefore, we may say that freedom is at the core of FSF philosophy – the freedom to use, study and customise, the freedom to redistribute, the freedom to cooperate. FSF philosophy is against software patents and additional restrictions in existing copyright laws. On the other hand, the OSI philosophy is slightly different from the FSF philosophy. The philosophy of OSI gives less emphasis on the ethical issues as proposed by FSF. It is directed towards the practical rewards of the distributed development process of open-source software. It targets the technical values of the participatory software development model for developing software and is more business- friendly than the FSF. But there are many common issues in these two philosophies of open-source software development, such as efforts against the proliferation of commercial software, software patents and making the software development process easy and user friendly. Richard Stallman, the father of FSF, rightly said that the Free Software Movement and the Open Source Movement are two political parties in the same community (Wong and Sayo, 2004).

4.4.2 Open Source Software (OSS)

On the other hand, open-source software is one where there is an underlying 'source code' made available under a licence. The developers and users have to adapt and modify it as per their needs.

However, the distribution terms of open-source software must comply with the following criteria:

- I. **Free Redistribution:** The licence given shall not disallow the other party from selling the software as a component of an aggregate software distribution containing programs from different sources. Furthermore, the licence will not be given for fee or royalty for such a sale.
- II. **Source Code:** The program must have a source code, and there should be a distribution in that source code.
- III. **Derived Works:** The licence must allow changes and derived works and permit them to be distributed under the same terms as the licence of the original software.
- IV. **The integrity of the Author's Source Code:** The license may restrict source code from being distributed in modified form only if the licence allows the distribution of 'patch files' with the source code to modify the program at build time.
- V. **No Discrimination Against Persons or Groups:** The licence must not discriminate against any person or group of persons.
- VI. **No Discrimination Against Fields of Endeavour.** The licence must not restrict anyone from using the program in a specified field.
- VII. **Distribution of Licence:** The right attached to the program must apply to all to whom the program is redistributed without the need to execute an additional licence by those parties.
- VIII. **The licence must not be Specific to a Product:** The rights attached to a program must not depend on the program's being part of particular software distribution.
- IX. **The licence must not Restrict Other Software:** The licence must not place restrictions on other software distributed along with the licensed software.
- X. **The licence must be technology-neutral:** No licence promotion may be predicated on any individual technology or style of interface.

4.4.3 Licensing of Open Source Software

Licensing issues related to open source software are complex. Open-source software may be released under a variety of different licenses. Open Source Initiative (OSI) reported the availability of more than 60 licenses and

categorised these licenses under eight categories (<http://www.opensource.org/licenses/index.html>). However, an in-depth analysis shows that there are only two primary types of licenses, and countless variants are based on these two widely adopted licenses. These two main licenses are the GNU (recursive acronym for GNU's Not Unix) General Public License (GPL) and the BSD-style licenses.

The GNU General Public License (GPL)

The key features of GPL are – i) user freedoms is ensured and protected; ii) source code is always available; iii) users are allowed to copy, distribute and modify original code; iv) any changes made to a GPL program by the distributor must also be licensed under the GPL; v) distributors may not place any non-GPL restrictions upon the users; vi) recipients of GPL software are granted the same rights as the original distributor; and vii) a commercial software company cannot take a GPL program, modify it and then sell it under a different, proprietary license.

BSD-style Licenses

BSD-style (Berkeley System Distribution) licenses are identical to the original license issued by the University of California, Berkeley. These are among the most permissive licenses and include key features like – i) attribution is given to the original license holder by including the original copyright notice in source code files; ii) no attempt is made to sue or hold the original licensor liable for damages; iii) software code available under BSD-style license can easily be incorporated into commercial applications; and iv) BSD-style licenses do not require the distribution of source code (after modification of original code). These two major licenses may be compared against the following features in the context of distributing open-source software –

Damages; (i) software code available under BSD-style license can easily be incorporated into commercial applications, and (ii) BSD-style licenses do not require source code distribution (after modification of original code). These two major licenses may be compared against the following features in the context of distributing open-source software –

	GPL Licensed	BSD Licensed
Must distribute source code	Yes	No
Must distribute user-created source code	Yes	No
User-created source code must be available under GPL	Yes	No
Proprietary Software linking possible	No	Yes
Compatible with GNU GPL	Yes	No*

*The original BSD license is not GPL compatible, but the modified BSD license is compatible with GPL.

4.4.4 Open Source and Open Standards

Library services have long depended on shared standards. Recently, one question has been attracting our attention: whether a specific standard is open or proprietary. A proprietary standard is characterised by the fact that it is owned by someone (individual or organisation) who puts restrictions on - or can put restrictions on - users' access and use. On the other hand, a completely open standard has the following properties:

- It is accessible and free of charge to all (i.e. there is no inequity between users, and no payment or other considerations are required as a clause of use of the standard);
- It remains accessible (i.e. owners will not limit access to the standard later on, i.e. afterwards); and
- All standard aspects are translucent, well documented, and freely available.

The W3C (2006) provides a set of six-pack criteria in defining Open Standards:

- transparency (the due process is public, and all technical discussions, meeting minutes, are archived and citable in decision making);
- relevance (new standardisation is started upon due analysis of the market needs, including requirements phase, e.g. accessibility, multilingualism);
- openness (anybody can participate, and everybody does: industry, individual, public, government bodies, academia, on a worldwide scale);
- impartiality and consensus (guaranteed fairness by the process and the neutral hosting of the W3C organisation, with equal weight for each participant);
- availability (free access to the standard text, both during development and at the final stage, translations, and clear IPR rules for implementation, allowing open source development in the case of Web technologies); and
- maintenance (ongoing process for testing, errata, revision, permanent access).

Software development, as a process, depends on standards (de jury/de facto or proprietary/open) in each step. Open standards provide the following advantages – 1) free to apply for any lawful purposes; 2) open and collaborative development process; 3) well documented and no chance of data loss due to technical obsolescence. The visible disadvantages of open standards are – 1) availability of only a few major players (e.g. Loc, IFLA

etc.); 2) lack of coordination between open standard initiatives and open source software developers; and 3) non-availability of open standards in many important facets of library activities (e.g. exchange of bibliographic and authority data). Some of the well known open standards used in different library-related software are – MARC 21 family of standards for resource description, MARC-XML as exchange format, OAI/PMH as metadata harvesting standard, SRU/SRW as standards for web-based distributed searching etc.

4.5 TYPES OF SOFTWARE

Computer software can be broadly split into two development models.

1. Proprietary software, and
2. Open-source software (OSS)

As the name suggests, proprietary software or ‘closed software’ restricts access to and use of the source code. As per “Wikipedia”, proprietary software has restrictions on using and copying it, usually enforced by a proprietor. The prevention of use, copying, or modification can be achieved by legal or technical means. Technical means include releasing machine-readable binaries and withholding the human-readable source code. Legal means can involve software licensing, copyright and patent law.

The term is used by the Free Software Foundation to describe software that is not free software or semi-free software. Technically, the term means software with an owner who exercises control over the software. Thus, it can be used for all software not in the public domain. However, the FSF uses the term to highlight that the owner is of prime importance, in contrast to “free software”, where computer users’ freedom is of prime importance.

Proprietary software, thus, in other words, means that which is owned by an individual or a company (the one that developed it), and thus there are major restrictions on its use, and its source code is therefore kept secret.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. How does open-source software differ from proprietary software?

.....

4.6 DESIRABLE SOFTWARE ATTRIBUTES

The debate over the merits of proprietary software and OSS has gained great momentum in recent years. However, the following attributes are desirable both in open source and proprietary software.

Reliability – how far a system is reliable without any disturbance?

Quality – number of euros in a fixed number of lines of code

Security – how flexible the software is to unauthorised actions, e.g. (virus)?

Flexibility – how easily the software can meet the needs of different devices?

Project Management – how organised it is?

Open Standards – if a document is created in one type of software, can it be readable in another.

Switching Costs – the expense for moving from one system to another

Total Cost of Ownership (TCO) – the total cost expense over the lifetime of the software.

User Friendliness – how easy is it adopted by people?

4.7 ADVANTAGE OF OPEN SOURCE SOFTWARE

The usual notion is that open source software is made available at a low cost, but this can be true in the case of proprietary software too. Therefore following are some of the advantages of open-source software.

Since the source code is available and one can modify it, it is important. This enables the unlimited timing and improvement of a software product. This also makes it possible to port the code to new hardware, adapt it to the changing conditions, and reach a detailed understanding of how the system works. Now experts understand that to increase an application's lifetime, and it must be available in a source form. It has been seen that binary- only application does not survive for more than ten years in an unmodified form. However, open-source software systems have survived for more than twenty years and are still widespread. Because of the availability of source code, one can also isolate bugs and fix them.

Technical motivations are common because there are different ways to perform work at times, and one may not decide which one is better. So if the two parties do not reach a consensus and the user base is quite large, the code then splits into two, but both continue development. If the reasons for the split comes to an end, and both the parties agree, then there may be a reunification. In other cases, a 'fork' is used to coordinate work.

E.g. the Linux kernel has two distinct codebases, one stable and one experimental. This way, one can introduce new and potentially dangerous technologies without disrupting the stable ones.

Another advantage of open-source software is that no per-copy fee can be asked for modified versions, and thus anyone can use the current code base to start whatever new projects. One can gather knowledge at a very low cost.

Another advantage of OSS is minimal conflicting priorities due to marketing pressures. This is because no single commercial entity is pushing for precise delivery dates or features that must be supported. Usually, OSS is made available to people only when it is ready and when the development team believes it is good enough for use.

Finally, OSS gives a new platform for democratic action. One can say that it provides a platform for democratic action because it is designed the way people need it, improvements are made as per their needs. Thus, the community's collective desire determines the overall direction of development and modification without compelling anyone to do that. Thus, public opinion is the backing force that leads to the progress of such software.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

Fill in the blanks:

- 1) With open-source software, there is always the possibility of creating an base if the recent one is perceived as wrongly managed.
- 2) The is the backing force that leads to the progress of such Open Source Software.

4.8 LEGAL ISSUE

4.8.1 Copyright

Software is protected using the copyright system. The same protection is given as on books, music or film, and the buyer of software has licenced the use of a copy of the product. Software is usually not purchased, but a licence is given to use it. Copyrights are used to protect computer software, but all do not use copyrights to have rights in the software. Some use copyrights so that no one may have any rights in software; there is a new word for it, 'copyleft'.

As we know, software, where source code and object code are freely available, is called 'open-source software. In copyleft open source software, source code and object code are freely available to be used, modified and improved without any changes. The term open-source software is often used in the sense of open source software that is copyleft.

4.8.2 Software Patents

Whereas copyright protects the software code from being copied, patents can prevent the innovative solution or effects of software from being copied. Now,

usually, the government grants the patent holder rights in return for sharing the information on how the technical result was achieved. The extent to which the software should be patentable is controversial. A key issue is whether the software has a 'technical effect' or is used for business.

The scenario in the US is that the software can be patented for business processes. In the EU currently 'business processes are not patentable.

4.9 OTHER SUCCESSFUL OPEN SOURCE SOFTWARE

- 1) Open office.org suite
- 2) Mozilla
- 3) Ximian

Office suite provides a bundle of software that is used in an office. The most popular office suite is Microsoft Office Suite. Open Office.Org suite is similar to it.

The software which permits one to access the Internet is called Web Browser. There is much such software - Opera, Internet Explorer etc. Mozilla is an open-source web browser, and Mozilla both can be operated in Linux and Windows.

Microsoft Outlook is an electronic personal information manager. It manages one's email, calendar, appointments etc. Ximian is also an electronic personal information manager. It is similar to Microsoft Outlook; however, it works on Linux only.

A good number of open-source alternatives are available for popular proprietary software - find a list of these alternatives in this link - <https://opensource.com/alternatives>.

There are plenty of services available on the Internet classified as open-source applications - like MySQL, Apache, Content Management Systems, email applications, data analytics tools, data visualisation tools etc. You find detailed information and examples in this link - <https://opensource.com/resources/projects-and-applications>.

Check Your Progress: 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Name a few open-source software.

.....
.....

4.10 APPLICATIONS OF OPEN SOURCE IN OTHER FIELDS

The principle of open source can be applied to a variety of other applications and software development. They are:

Markets: Software is not the only field affected by open source; many fields of study and social and political views have been affected by the growth of the concept of open source. Advocates of one field will often support the expansion of open source in another field, including Linus Torvald, who is quoted as saying, “the future is open source everything”.

The open-source movement has inspired increased transparency and liberty in other fields, including the release of biotechnology research by CAMBIA and the encyclopedia named ‘wikipedia’.

Cola Drinks: Open cola is another idea inspired by the open-source movement. Soft drink giants like Coke and Pepsi hold their formulas closely guarded secrets. Now volunteers have ported the recipe for a similar soda drink on the Internet. The taste is said to be comparable to that of the standard beverages.

Beer: There is also an open-source beer called Varese.

Pharmaceuticals: There have been several proposals for open-source pharmaceutical development, like the one that led to establishing the tropical disease initiative. There are also several not for profit “virtual pharmas” such as the Institute for one World Health and the Drugs for Neglected Diseases Initiatives.

Let us now summarise the points covered in this Unit.

4.11 LET US SUM UP

- Open-source software, in other words, means software where the source code is made available to all, and a licence is given for the modification and development of the software as per the user’s need.
- The availability of the source code makes it simpler to use by the user.
- However, there are legal issues related to such software: copyright and patent issues.
- The open-source movement is gaining momentum in other fields other than the software, i.e. we now have open colas, open-source in software, hardware, pharmaceutical industries etc.

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4.13 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. In the late 70's and early 80's Richard Stallman, an American software developer who had an idea that sharing source code and ideas are fundamental to freedom of speech developed a 'free' version of the widely used 'UNIX' operating system. This resulted in a 'GNU' program released under a specially created General Public Licence ('GNU.GPL'). The GPL is the license under which much Open Source code is distributed. It enshrines the idea that computer code, like speech, should be free to be copied, interpreted, modified and generally mashed up. This was designed so that source code could remain openly available to all. This software was designed for commercial usage or distribution. Such an approach was called 'free software. Thus the term “free” denotes that anyone could modify the software as per his/her needs.

Check Your Progress: 2

1. As the name suggests, proprietary software or ‘closed software’ restricts access to and use of the source code. As per “Wikipedia”, proprietary software has restrictions on using and copying it, usually enforced by a proprietor. The prevention of use, copying, or modification can be achieved by legal or technical means. Technical means include releasing machine-readable binaries and withholding the human-readable source code. Legal means can involve software licensing, copyright and patent law. The term is used by the Free Software Foundation to describe software that is not free software or semi-free software. Technically, the term means software with an owner who exercises control over the software. Thus, it can be used for all software not in the public domain. However, the FSF uses the term to highlight that the owner is of prime importance, in contrast to “free software”, where computer users’ freedom is of prime importance.

Check Your Progress: 3

1. Alternative code
2. Public opinion.

Check Your Progress: 4

1. Mozilla, Linux.

UNIT 5 NEW MEDIA AND ETHICAL ISSUES

Structure

- 5.0 Introduction
- 5.1 Learning Outcomes
- 5.2 Definition of New Media Ethics
- 5.3 Rights and Ethical Responsibilities of Content Creators
 - 5.3.1 Understanding Digital Rights and Creative Commons
 - 5.3.2 Ownership Issues
 - 5.3.3 Copyright Infringement
 - 5.3.4 Open Content vs Free Content
- 5.4 Content Curation and Limits to Sharing
 - 5.4.1 Privacy and Other Ethical Issues
 - 5.4.2 News Related Ethical Issues
 - 5.4.3 Ethics for Online Marketers
- 5.5 Rights and Ethics of Online Readers
 - 5.5.1 Instances of Ethical Violations
 - 5.5.2 Do's and Don't's of New Media Ethics
 - 5.5.3 Dealing with Ethical Violations
- 5.6 Let Us Sum Up
- 5.7 Further Readings
- 5.8 References
- 5.9 Keywords
- 5.10 Check Your Progress: Possible Answers

5.0 INTRODUCTION

The advent of new technology has opened up a world of information for us. Internet, the World Wide Web, Smartphones, mobile applications etc., have made us familiar with terms such as New Media. Earlier, the distinction between new and old or traditional media was clear, but the constantly evolving technology has forced us to rethink the definition of new media. We have come a long way from a one-dimensional definition of the concept. New media is a broad term that carries multiple meanings and interpretations. Using new technologies and advanced devices for communication has created a scenario where we are constantly consuming online content. Today we spend much more time being 'logged on' than we have ever done.

New media is not only changing the way we interact with our surroundings; it is also dictating 'how' we do it. It facilitates sharing in a number of ways,

including text, video, pictures, graphs, illustrations, audio etc., asking each user to engage with the online content more deeply and uniquely. The user has to be able to make sense of all this information and understand the pitfalls of communicating online.

There is a range of activities such as emailing, video sharing, social networking, virtual gaming, and even hacking that can be done online. There are unseen attacks such as hacking, phishing and trolling that make our experience of the online world uncomfortable. Law means legal rules used to govern a country's matters. The Indian Penal Code defines the Indian penal system, and the Information Act 2000 is the guiding principle for new media in India. The IT Act and its subsequent amendment in 2008 take care of defamation, libel and slander apart from more serious online crimes such as phishing and fraud. But the huge amount of data uploaded cannot be controlled by laws alone, and the need of the hour is to understand new media ethics.

5.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- describe the different aspects of new media ethics;
- discuss the importance and challenges related to it;
- explore the responsibilities of users while accessing or downloading content; and
- use new media with greater care in your personal and public online conduct.

5.2 DEFINITION OF NEW MEDIA ETHICS

New Media ethics define how we behave online and help us decide about our 'netiquettes'. These are required because the information is being exchanged quickly and in huge amounts. Both consumers and producers of new media need to be aware of their ethical limitations in the online world. While there are legal rights to protect content creators, the policies for the consumers of online content are in nascent stages.

A great deal of content is being created online. It is important that the Internet remains open, transparent and protects the rights of individuals who contribute to the sharing and exchange of knowledge. This allows them to create content freely and fosters creativity while making knowledge accessible to everyone. The temptation to copy or modify easily available content is difficult to fight. The ownership, privacy, confidentiality and intellectual property need to be respected in virtual life just like in our real life.

Therefore new media ethics can simply be described as a set of ethical considerations that define our conduct online. New media ethics tell us how

to appropriately use new media without breaching any law or infringing on the online rights of other users. They provide a model code for creators, distributors and users of online content.

Collaborative websites such as Wikis and YouTube, social media sites such as Facebook, MySpace, Weibo and Twitter are all running with the help of user-generated content. Individuals who are reading the content also share, exchange, add-on and publish the data they come across and thus end up becoming the consumer and producers. This is unlike any other medium, e.g. we use Facebook to share pictures and 'like' pictures found on our friends' walls, thereby creating data. Thus, new media users need to be protected from ethical violations as producers and creators of information. Let us first try to understand their responsibilities as creators.

5.3 RIGHTS AND ETHICAL RESPONSIBILITIES OF CONTENT CREATORS

The Copyright Law protects individuals from having their work copied in an unauthorised way. In contrast, the Information Technology Amendment Act 2008 has a small section dedicated to dealing with cyber crimes such as sharing obscene material or impersonating on the Internet. Since the laws have not evolved as fast as the medium itself, it becomes imperative to follow the ethical guidelines strictly.

5.3.1 Understanding Digital Rights and Creative Commons

Digital Rights have become very important in this information age. The rights of both the content creator and consumer or reader need to be protected. There are many challenges here as consumers on the Internet are mostly prosumers (producer + consumer). They have to watch out for ethical violations both as creators and consumers of online content. Since establishing ownership is easier, the content creator's rights are always discussed more fervently. New techniques collectively referred to as Digital Rights Management (DRM) have evolved to protect these. They make it difficult to duplicate access or distribute data in an unauthorised way. We see some common techniques in everyday life: copy control, access control, and digital watermarking. When we use a watermark, there is a link to the original work embedded somewhere within the text to guide the reader to the original work. One can see websites where only a part of the information is free while one needs to be either a member or a paid subscriber to access the text, such as Google Books, SAGE etc. Some software controls are also designed to disable any copying and thereby automatically prevent data reproduction. More sophisticated techniques involve data encryption. The data is jumbled up and pieced together only by authorised persons with the decryption key, much like a jigsaw puzzle.

The major issues with respect to new media ethics for content creators are:

- Ownership and Copyright

- Open vs Free content and Content sharing
- Privacy and other ethical issues

5.3.2 Ownership Issues

The web shows information readily and what we need is available at one click. This is also true for pictures, music, graphics, songs and any other information that we seek. But that also throws up a concern about protecting our rights as creative people. Though the Copyright Law legally protects us, we are ethically bound by ownership and copyright issues on the web. There are new forms of copyright available online now.

5.3.3 Copyright Infringement

There are a lot of websites offering paid content online. This is done to ensure that artists/writers contributing content are compensated and motivated to contribute further. But some file-sharing companies allow illegal downloading and sharing of files that lead to piracy. The rampant music piracy or copying of digital goods such as audio files, videos, reports, articles and such without the consent of the person who holds the copyright is a major problem. As users, we tend to become careless about copy-pasting desired information and infringing upon fellow netizens' copyright.

5.3.4 Open Content vs Free Content

David Wiley has specified that online content can be both open and free. It is important to distinguish between the two. Any content available online that can be accessed, revised, reused, redistributed, remixed and retained for later use is referred to as open content. This means that the online content need not be 'open' even if it is free, implying that free content is not the same as 'open' content. Because free means one can access the content free of cost, but 'open' means users are free to download, edit and share the content without any ethical or legal limitations. Free content allows individuals to use and benefit from it, make and distribute its copies, study and apply it or even distribute its derivative works.

The problem of creating ethical guidelines for the Internet has risen because it is a relatively new medium. There is a vast amount of content floating around on the Internet, and there are many conventions that define our ethical boundaries with respect to using the content available online. The automatic copyright of the content posted online lasts for 70 years, after which it becomes available for public use. But new types of copyright licensing available online now tell us how a particular content should be used. Creative Commons is one such organisation working on the idea of providing people access to creative work and intellectual property while giving them a right to choose the conditions of making that work available to consumers. In simple terms, whenever a user puts up any fresh piece of information online, he/she can specify the way it has to be used by mentioning if the material is free for

modification/download/sharing etc. or partly available/ not available. This is done via symbols and attributions that have now come to be used commonly on the Internet.

These guidelines are helpful in understanding how one must use content available online. However, it should be noted that licenses issued by Creative Commons are not a substitute for copyright. The table as an Annexure of this Unit explains the different types of copyrights from Creative Commons.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Define new media ethics.

.....
.....

2. What does Digital Rights Management mean?

.....
.....

3. Distinguish between Open and Free Content.

.....
.....

5.4 CONTENT CURATION AND LIMITS TO SHARING

The content writing style often keeps changing, and these days the major emerging trend is that of content curation. It is defined as a process of sorting, arranging and publishing data related to a specific theme together so that the reader gets more information about the same topic without having to click through multiple sites.

The work of curators is to pick the most relevant content and exhibit it so that a quick glance reveals the most relevant links at one place itself. It is unlike a search engine because here, the curator adds some inputs about the content to give the reader an idea about what the text carries and why they should click it.

There is a thin line between highlighting someone's content and presenting it as your own. While curating the content, the appropriate links and credits must be provided along with the text. Though giving an embedded link to the original text is within permissible copyright limits, it could be misleading for the readers, causing an ethical violation. Therefore, the curators must make strong efforts to mention the original source and links explicitly.

5.4.1 Privacy and Other Ethical Issues

The ethical constraints of content producers are also limited to the extent that they should not upload, distribute, share or misuse any content such as photographs, videos, music, articles, books, or any intellectual property acquired physically or virtually without the consent of the individual concerned. However, we often find a rampant violation of this basic new media ethic. Personal information of friends, relatives and acquaintances is often shared online without seeking explicit approval of the people in question.

5.4.2 News Related Ethical Issues

All the content on news websites may not come from reliable sources and can also be inaccurate. In such cases, it is prudent that the users be informed when a news item is less reliable or has only been edited after taking from another source. This helps them to be better prepared to handle such information. In the case of images, new end technologies have facilitated easy capturing, sharing and transmission of data. Also, in many cases, the news may not be verified, or wrong images could appear besides stories, thus requiring the reader to be discerning.

The reports generated through anonymous sources and hoax data are also commonplace. During the second season of IPL, a blog named 'Fake IPL Player' was started. The anonymous writer was believed to be a Kolkata Knight Riders team member but could not be identified. Only after the writing caught international attention and garnered a huge number of visitors did the blogger's identity get revealed. Before that, the team management had taken measures like curbing internet access and banning laptops from stopping the posts. This is the downside of information coming in through new media. It is impossible to ascertain the source and authenticity of the information correctly.

The problem is also faced by news employees who want to share their thoughts online. The companies are often not open to allowing this since it can lead to all kinds of problems. For instance, remarks of a person about a group or community could be hurtful and incite violence ending up spoiling the company's reputation. To avoid this, many companies have now come up with clear guidelines so that their employees can use social media fearlessly. They often spell this out as 'Social Media Policy'. The leading publication Los Angeles Times has drawn up a social media policy for this very reason. It says that:

1. Principles of integrity, professionalism, privacy and impartiality should be observed by journalists when posting online.
2. The authenticity of what employees post is important. Online journalists should verify questionable content with credible sources before posting or tweeting about it.

3. It is important for employees to properly define their association with the publication as they would do offline.

Some companies such as Adidas go one step ahead and give clear instructions to the employees, as outlined below:

1. Employees are allowed to associate themselves with the company when posting, but they must clearly brand their online posts as personal and purely their own. The company should not be held liable for any repercussions the employees' content may generate.
2. Content pertaining to sensitive company information (particularly those found within Adidas internal networks) should not be shared with the outside online community. Divulging information like the company's design plans, internal operations and legal matters are prohibited.
3. Proper copyright and reference laws should be observed by employees when posting online.

Thus it is clear that both content producers and readers need to have a system of checks and balances for their online behaviour. Their etiquettes cannot be controlled but can be guided with the help of social media policies, and drawing up ethical guidelines etc.

5.4.3 Ethics for Online Marketers

Since users are increasingly connected with each other and keep sharing intimate information online, the marketers often reach them through advertisements that are specifically targeted as per their browsing history. This is done through cookies that track the data of users. They not only spoil the experience of the users but are considered unethical. The use of hidden hyperlinks and contextual link ads is very annoying. There are also instances of companies floating paid reviews and sponsoring content that is favourable for them. This creates positive reviews for the product but is a gross violation of ethics since it is misleading for the readers. Sometimes malicious content appears in the form of pop up ads/ alerts, in-app clicks or download links. They may turn out to be viruses. The users need to take special care when dealing with such content.

Search engine optimisation is also used as a technique to gain users' attention. However, this is used indiscriminately to redirect users to the same sites repeatedly.

As per the **Institute of Advertising Ethics (IAE)**:

- "All forms of advertising material must share the common goal of maintaining the truth and should be a means to serve the public.
- They should maintain a clear distinction between corporate communications, press releases, sales collateral and advertisements.

- A publisher must disclose every condition upfront and clearly, as the asterisks (*) and fine print at the end of the document can sometimes go unnoticed by the consumer.
- If cookies are being used to track and detect a user's settings, personal record and online activity, then this must be clearly stated before the user begins browsing your site.
- The placement of ads should not obstruct the user view, nor should they be disguised as editorial content.
- Finally, whether you are selling online or offline, advertisers must abide by the federal, state and local advertising laws.”

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What are the responsibilities of content creators in the online world?

.....
.....

2. How can one become more alert towards personal data being published online?

.....
.....

3. Why is social media policy important in the present context?

.....
.....

5.5 RIGHTS AND ETHICS OF ONLINE READERS

It is not just the content creators who need to be restricted by ethical boundaries; there are unwritten protocols for readers. Something as innocent as ‘liking’ a page or tweeting on a trending topic can sometimes lead to a violation of online ethics due to the nature of the content. Other issues facing readers are cyber disinhibition, inadvertent self-disclosure, cybercrimes such as bullying, doxing, phishing etc. On the consumers’ end, especially social media users, the invasion of privacy is not fully understood.

Readers in the online world are still unaware of the data trails they willingly give into the hands of internet service providers and are thus falling into a situation where they have no control over their own data. The disclosure problem is compounded by the lack of awareness among Internet and new media users, which can lead to invasion of privacy.

The companies often dictate the terms of service and privacy policies of social networking sites as per their requirement, and readers have little or no say in changing them. They are also unaware of their virtual identities and how data-mining firms are being used.

Similarly, social networking sites often collect information and share it with third-party users. Still, users cannot control this since failure to accept these policies often results in being unable to sign up or access these sites and applications.

Thus, we reveal much more about ourselves online than we had planned to simply because we do not care to go through the privacy settings or disclosure agreements while downloading apps or accessing social media. The least a user can do is read and understand these carefully and limit the amount and type of personal information they upload. The increasing cyber disinhibition or the loosening of social restrictions and inhibitions in the virtual world is a major cause of ethical and legal violations.

Activity – 1

Check the privacy settings of the social networking site of your choice and list its salient features.

5.5.1 Instances of Ethical Violations

The new media is a powerful tool to reach out to people, but it is sometimes misused for personal attacks, running malicious campaigns with intended or unintended consequences. Trolling, cyber vandalism and smear campaigns are mounted on the Internet to garner quick support.

Sometimes these incidents go out of hand and create gross ethical and legal violations. For example, the pictures of the brutal murder of a northeastern student in the Munirka area of Delhi were shared online without any respect for the privacy of the individual or his family's consent. Similarly, leaked pictures of film stars holidaying abroad caused a lot of annoyance to the persons concerned. In the above cases, we can hold the person who circulated the pictures legally guilty, but people who came across the pictures and shared them further indulged in gross ethical violations. In case any content being circulated on the net is found to be objectionable, it must be reported to the particular site and the authorities so that necessary action may be taken.

Another ethical violation is the trend of shaming videos. These are videos where people are humiliated with punishments such as chopping off the hair, beating etc., as punishment and are recorded to be uploaded online with the motive of promoting online embarrassment. There have been reports of kids, teenagers and even adults feeling suicidal after their shaming videos came online. This disturbing trend also includes videos of pet shaming. Though they are intended to serve as a punishment, such videos often go 'viral' and can affect the mental and social health of persons concerned.

People of all age groups use the Internet, and not all users are discerning enough to understand the consequences of the data they may share online. Anything once posted on the Internet remains there forever and can be dug up any time later and cause embarrassment. Therefore we must be extra careful in putting up or sharing our pictures with friends and colleagues.

We often find the abusive and derogatory language used on web page comment sections, chat rooms and social networking sites, etc. This type of trolling results from disinhibition or breaking down of socio-cultural norms in the virtual world. Another big threat is the addiction to these sites. The overuse of apps and social networking sites is causing an alarming situation where the addicted users are psychologically more dependent on their virtual identities than on real-world scenarios. The ubiquitous 'selfie' also becomes a menace at times. Clicking a selfie at any and every minute of the day is not a crime. However, how, where and when it is clicked decides whether there has been an ethical violation or not. The former U.S. President Barak Obama faced a lot of flak for clicking a selfie at the funeral of Nelson Mandela.

As we look at the advent of the 'Internet of things, i.e., devices connected with the Internet that can be remotely operated, the responsibility of using the technology in a better way has also increased manifold.

5.5.2 Do's and Don't's of New Media Ethics

The guidelines for 'netiquette' are very simple. When we post something in relation to a text, we should point to the original text while commenting so that our post is not taken out of context. It is important that we are careful while posting any data online, and before doing it, one must consider the long term consequences.

We should be sensitive to the online readers who may represent different beliefs, sensibilities and cultures. If we have committed a mistake, we must immediately apologise and remove the concerned posted text. We should be gracious in disagreements too. Expressing any negative opinion online on a public forum can disturb any individual. Below is a list of things one can do to prevent unethical practices online.

Do's and Dont's of New Media Ethics

- "Do not open messages from strangers.
- Do not forward messages/rumours without first verifying the claims.
- Do not impersonate individuals or secure their passwords through unauthorised means.
- Never send pictures to strangers.
- Avoid downloading copies of pirated software or movies, songs etc.
- Respect the privacy of others, just as you expect the same from others.
- Do not use other users' computer resources without their permission.

- Use the Internet ethically.
- Complain about illegal communication and activities, if found, to Internet Service providers and local law enforcement authorities.
- Users should not intentionally use the computers to retrieve or modify the information of others, which may include password information, files, etc.”

5.5.3 Dealing with Ethical Violations

There are many pitfalls as far as ethical violations are concerned. Users may end up causing a lot of these and may choose to remain ignorant about them. For example, piracy of movies, TV shows and songs is very commonplace. People who download files through torrents or other means are well aware that they are causing ethical violations, but they do not think of it as a wrong practice. The Internet, which has facilitated many activities, has also heightened possibilities of falling into a ‘cut-copy-paste’ trap.

The foremost thing is that we should double-check the data we post online and be critical, observant and watchful while sharing content or talking to strangers. We should also encourage our acquaintances and friends to point out any inadvertent errors or mistakes we might have committed. As users of new media, we need to understand the constantly changing online world and figure out how to deal with emerging ethical challenges. We also need to draw up policies to ensure no conflict of interest between content creators and consumers online. The users need to be more aware of new crimes being committed online. Above all, we need to be aware of new media ethics and be willing participants in making the Internet a more productive, responsible and decent place for all users.

Activity – 2

Identify a recent instance of ethical violation on the Internet and discuss how it could have been prevented.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. What can readers do to safeguard their online experience?

.....
.....

2. How can online ethical violations be prevented?

.....
.....

5.6 LET US SUM UP

In this unit, we started with the definition of New Media ethics which were described as ethical considerations that define our conduct online that help us interact with people. We then moved on to the rights and responsibilities in the online world in which it was explained that there are ethical considerations both as creators and receivers of information. The legal part is taken care of by Digital Rights Management, Copyright laws and Intellectual property rights. However, the ethical part is a lot more complex and is dealt with by a combination of different mechanisms such as Creative Commons and Social media policies. An important aspect of understanding online ethics and using content over the web is to learn to differentiate between 'open' and 'free' content. These were discussed in detail, along with some ethical violations and how these can be handled effectively.

Some guidelines for new media use and a roadmap for online conduct were outlined in the unit. Through the discussion, a strong need for awareness among 'netizens' regarding their rights and responsibilities in the online world and social media policies were emphasised.

It was stressed that new media users need to be careful about their personal and public conduct in the online space. This can be achieved by understanding the ethical constraints of new media users.

5.7 FURTHER READINGS

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5.9 KEYWORDS

New media ethics	: Ethical guidelines for online behaviour.
Netiquettes	: Convention for communicating in cyberspace is made from the word etiquette colloquially meaning ‘good manners’.
Wikis	: Open and Collaborative content that is user-generated and can be edited by anybody.
Cyber Disinhibition	: the phenomenon of breaking down of social and moral codes of conduct in the virtual world.
Content Curation	: It is the manual sorting and arrangement of data available online so that the reader gets all related text in one place itself. This saves the reader time and provides customised yet varied content to readers without going through various links and websites.

5.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. New Media Ethics are ethical considerations that define users’ online conduct. These help the consumers appropriately use new media technologies without infringing the rights of other users or breaking any laws.
2. Digital Rights Management (DRM) has evolved to protect the rights of online content creators. This help prevents the unauthorised duplication,

access and distribution of data. Copy control, Access Control, and digital watermarking are some of the ways used for this.

3. Open content is online content that can be accessed, revised, reused, redistributed, remixed and retained for later use. At the same time, free content can be accessed without paying a fee but cannot be edited, revised or reused further.

Check Your Progress: 2

4. While using content, one should provide appropriate links and credits along with the text and explicitly mention the source. They should also not upload, distribute, share or misuse any content such as photographs, videos, music, articles, books, or any other intellectual property acquired physically or virtually without the consent of the individual concerned.
5. Personal information of friends, relatives and acquaintances is often shared online without seeking their explicit approval. One must respect the privacy and copyright of others while publishing content online.
6. Social media is designed to be an easy-to-use system for voicing our opinions. But there is a need to be aware of the new kind of dangers it exposes us to. Social media policies define the limits of actions that can be performed on social media and thus help us follow responsible online behaviour - both as content producers and users.

Check Your Progress: 3

1. Social networking sites often collect information and share it with third party users. Although users cannot control this, it is prudent to go through the privacy settings or disclosure agreements while downloading apps or accessing social media. The user should read and understand these carefully and limit the amount and type of personal information they upload.
2. We should double-check the data we post online and be critical, observant and watchful while sharing content or talking to strangers. We should also encourage our acquaintances and friends to point out any inadvertent errors or mistakes we might have committed. Copyrights granted by the Creative Commons.

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